

Memoir

On the impossibility of some indeterminate equations of the fifth degree

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by G. Lejeune Dirichlet.

According to the report of Messrs. Lacroix and Legendre, this memoir was approved and is to be printed in the
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It is known that the theory of indeterminate equations of degrees higher than the second is still very little advanced. It is true that there are infinitely many equations of all degrees whose impossibility can be proved by showing that, whatever values are assigned to the indeterminates, the two sides of the proposed equation can never give the same remainder when divided by a certain number, or modulus. But when an equation cannot be treated by this means, it becomes difficult to prove that it is impossible; up to the present this has been done only for a very small number of equations. All these equations are very special, and of such a form that, when one attempts to solve them, one is naturally led to one or more quadratic formulae which have to be equated to perfect powers. One then satisfies this condition in the most general manner by expressing the indeterminates by other indeterminates, of which the first become integral functions; and it turns out, at least in all the cases in which the method in question succeeds, that the new indeterminates, or other quantities depending on them in a very simple manner, likewise satisfy an equation similar to the proposed equation. Since the new indeterminates are at the same time smaller than the primitive indeterminates, the impossibility of the proposed equation is established; for it is evident that, if it were possible, one would have a means of obtaining a decreasing and indefinite sequence of integers, which is a contradiction. In this manner Fermat and Euler proved the impossibility of several equations of the third and fourth degrees.

In trying to apply similar considerations to some equations of the fifth degree, and of a form analogous to those of the equations treated by Fermat and Euler, one is stopped at once. The quadratic formula at which one arrives, and which has to be equated to a fifth power, admits several different solutions, and among these solutions there is only one which leads to an equation similar to the proposed equation. Reflecting on this difficulty, I recognized that it could be removed very simply by subjecting the fixed number which enters the equation to certain conditions. It follows from a theorem stated in the preliminaries that, when these conditions are fulfilled, the different solutions of which the quadratic formula is in general susceptible must be rejected, with the exception of one alone, which is precisely the one from which one deduces numbers satisfying an equation similar to the proposed equation. In this way one arrives at the impossibility of a fairly extensive class of indeterminate equations of the fifth degree. The first side of these equations is the sum or the difference of two fifth powers, and the second side is the product of a fifth power and of a fixed number subjected to various conditions. By assigning to this number particular values compatible with these conditions, one may obtain as many particular theorems as one wishes. This generality of our theorems is all the more striking because the analogous equations of the third and fourth degrees, whose impossibility has so far been proved, are only finite in number and even very few.

The numbers P and Q being coprime, one even and the other odd, and the first moreover not divisible by 5, it can be proved that, in order to equate the binomial $P^2 - 5Q^2$ to a fifth power with

all suitable generality, one need only put

$$P + Q\sqrt{5} = (M \pm N\sqrt{5})^5(t \pm u\sqrt{5}),$$

the rational parts and the coefficients of $\sqrt{5}$ being equated separately, t and u satisfying in general the equation $t^2 - 5u^2 = 1$, and the numbers M and N being coprime, one even and the other odd, and the first not divisible by 5¹. This supposed, it is not difficult to establish the theorem which we are about to state.

Theorem I.

The numbers P and Q being coprime, one even and the other odd, and the latter moreover divisible by 5, I say that, in order to equate the binomial $P^2 - 5Q^2$ in the most general manner to a fifth power, it will suffice to put

$$P + Q\sqrt{5} = (\varphi + \psi\sqrt{5})^5.$$

The indeterminates φ and ψ are to be coprime, one even and the other odd, and the first moreover not divisible by 5².

To equate $P^2 - 5Q^2$ to a fifth power, we shall put, according to what has just been said,

$$P + Q\sqrt{5} = (M \pm N\sqrt{5})^5(t \pm u\sqrt{5}).$$

Since N is not divisible by 5, if for a moment we set

$$(M + N\sqrt{5})^5 = M' + N'\sqrt{5},$$

it will be easy to see that N' is divisible by 5 and that M' is not. Substituting the preceding expression into the value of $P + Q\sqrt{5}$, we obtain

$$P + Q\sqrt{5} = (M' \pm N'\sqrt{5})(t \pm u\sqrt{5}),$$

whence one draws

$$Q = \pm M'u \pm N't.$$

Since N' is divisible by 5 and M' is not, it is evident that Q can be divisible by 5 only insofar as u is. The smallest values satisfying the equation

$$t^2 - 5u^2 = 1$$

are these:

$$t = 9, \quad u = 4.$$

The general values will therefore be given by this formula,

$$t + u\sqrt{5} = (9 + 4\sqrt{5})^p,$$

¹In order not to give this memoir too great an extent, I suppress here a first part of the manuscript memoir containing some theorems on numbers insofar as they are of the form $t^2 - au^2$, and the proof of the proposition in question founded on those theorems.

²It may not be useless to remark that there are analogous theorems for many other primes, and that, to establish them, one may use the same considerations which we use here.

in which p is any positive integer³; from this one obtains

$$u = \frac{p}{1} 9^{p-1} \cdot 4 + \frac{p(p-1)(p-2)}{1 \cdot 2 \cdot 3} 9^{p-3} \cdot 4^3 \cdot 5 + \text{etc.}$$

All the terms of this value, beginning with the second, being divisible by 5, whatever p may be, one sees that, in order that u may be divisible by 5, the first term, and therefore also p , must be divisible by 5. If therefore we put $p = 5p'$, p' being an integer, and substitute the value of $t + u\sqrt{5}$ in that of $P + Q\sqrt{5}$, we shall have

$$P + Q\sqrt{5} = (M \pm N\sqrt{5})^5 (9 \pm 4\sqrt{5})^{5p'},$$

a result which, by introducing the new indeterminates φ and ψ defined by

$$(M \pm N\sqrt{5})(9 \pm 4\sqrt{5})^{p'} = \varphi + \psi\sqrt{5},$$

changes into

$$P + Q\sqrt{5} = (\varphi + \psi\sqrt{5})^5.$$

The form of the solution given by the statement of the theorem being thus justified, it remains only to determine the nature of the indeterminates.

Since one has

$$P^2 - 5Q^2 = (\varphi^2 - 5\psi^2)^5,$$

and since the numbers P and Q are respectively divisible by φ and ψ , one sees easily that the indeterminates φ and ψ must be supposed coprime, one even and the other odd, and the first moreover not divisible by 5. One may even add that φ or ψ will be odd according as P or Q is. Conversely, if the indeterminates φ and ψ satisfy the preceding conditions, the numbers P and Q determined by the formula

$$P + Q\sqrt{5} = (\varphi + \psi\sqrt{5})^5$$

will be coprime, as is easy to verify. These preliminaries established, we may turn to the theorems which are the principal object of this memoir. The first of these theorems may be stated as follows.

Theorem II.

Let the numbers m and n be positive, greater than zero, the second moreover different from 2, and let the number A be divisible neither by 2 nor by 5, nor by any prime number of one of the forms $10k \pm 1$ ⁴. Then it is impossible to find two coprime numbers x and y such that

$$x^5 \pm y^5 = 2^m 5^n A z^5. \tag{\alpha}$$

³See the Additions to Euler's Algebra, article 75, or the *Disquisitiones arithmeticae*, article 200.

⁴The theorems of this memoir, like those contained in the addition, admit an extension which I shall indicate in a few words. These theorems still hold even when A has prime divisors of the form $10k - 1$. Indeed one will see that our proof assumes only that the number A have no common divisor with the formula $t^4 + 10t^2s^2 + 5s^4$, where t and s are supposed coprime. Now, this formula can be put in the form $\frac{(t+s)^5 + (t-s)^5}{2t}$; and, since 5 is prime, it follows from Euler's known theorems on the linear forms of the prime divisors of the expression $x^n + y^n$ that our formula has only prime divisors of the form $10k + 1$, and is consequently prime to A . See the Theory of Numbers or Euler's memoir *circa divisores numerorum*, in the first volume of the *Novi Acad. Comment. Petrop.*

Suppose, contrary to the statement of the theorem, that the equation is possible. Since the second side is even, m having been supposed > 0 , the numbers x and y , which are coprime, must both be odd. If we put $x \pm y = 2p$, $x \mp y = 2q$, and consequently

$$x = p + q, \quad \pm y = p - q,$$

the numbers p and q will be integers, coprime, and moreover one of them even and the other odd. Substituting the preceding values of x and $\pm y$ in equation (α) changes it into

$$2p(p^4 + 10p^2q^2 + 5q^4) = 2^m 5^n A z^5.$$

The first side can be equal to the second, which is divisible by 5, only if p is supposed divisible by 5. Thus putting $p = 5r$, we have

$$2 \cdot 5^2 r(q^4 + 2 \cdot 5^2 q^2 r^2 + 5^3 r^4) = 2^m 5^n A z^5.$$

By hypothesis the number n is either equal to unity or greater than 2. If n is equal to unity, it will be necessary in the preceding equation to suppose z divisible by 5. In that case one may put $5z$ in place of z , or, what is the same thing, give to 5 the exponent 6; hence one sees that in all cases one may suppose that $n > 2$. If now the preceding equation is put in the form

$$r(q^4 + 2 \cdot 5^2 q^2 r^2 + 5^3 r^4) = 2^{m-1} 5^{n-2} A z^5,$$

and one remembers that the numbers q and $p = 5r$ are coprime and moreover one even and the other odd, it will be easy to see that the trinomial factor is odd, not divisible by 5, and prime to r ; therefore r must be divisible by 5, since $n > 2$. Choose now two positive numbers μ and ν such that $m + \mu - 1$ and $n + \nu - 2$ are divisible by 5, and a number B having no prime divisors other than those of the number A , and such that the product AB is a fifth power. If we multiply the preceding equation by $2^\mu 5^\nu B$, we obtain

$$2^\mu 5^\nu B r(q^4 + 2 \cdot 5^2 q^2 r^2 + 5^3 r^4) = 2^{m+\mu-1} 5^{n+\nu-2} A B z^5.$$

The second side of this equation being a fifth power, the first side will likewise be one. I claim that the two factors into which this first side may be decomposed, namely the factor $2^\mu 5^\nu B r$ and the trinomial factor, are coprime. Indeed, we have already seen above that the trinomial factor is prime to $2^\mu 5^\nu r$, and, to make sure that it is also prime to B , we put it in the form

$$(q^2 + 5^2 r^2)^2 - 5(10r^2)^2.$$

The numbers $q^2 + 5^2 r^2$ and $10r^2$ being evidently coprime, every prime by which the trinomial factor is divisible will be of one of the two forms $10k \pm 1$, neither of which, by hypothesis, is suitable for the prime divisors of A , and consequently also of B , since B has no prime divisors other than those of A . Thus the number $2^\mu 5^\nu B r$ and the trinomial factor must each be fifth powers.

The trinomial factor can be written in this manner,

$$(q^2 + 5^2 r^2)^2 - 5(10r^2)^2,$$

and the numbers $q^2 + 5^2 r^2$ and $10r^2$ are coprime, the first odd and the second even and divisible by 5. Hence, by Theorem I, in order to equate the trinomial factor, with all suitable generality, to a fifth power, it will suffice to put the two equations

$$q^2 + 5^2 r^2 = t(t^4 + 2 \cdot 5^2 t^2 s^2 + 5^3 s^4),$$

$$10r^2 = 5s(t^4 + 10t^2s^2 + 5s^4).$$

The numbers s and t must be supposed coprime, and moreover the first even and the second odd and not divisible by 5. It follows that s must be divisible by 5: indeed, since t is not divisible by 5, the second side of the second equation cannot be divisible by 5^2 unless s is divisible by 5; but the first side, which has r^2 as a factor, is divisible by 5^2 ; therefore s is divisible by 5.

We have seen above that $2^\mu 5^\nu Br$ had to be a fifth power. The number $2^{2\mu} 5^{2\nu} B^2 r^2$, the square of the preceding number, must therefore be a power of the same degree, and indeed a tenth power. Now, multiplying both sides of the last equation by $2^{2\mu-1} 5^{2\nu-1} B^2$, we obtain

$$2^{2\mu} 5^{2\nu} B^2 r^2 = 2^{2\mu-1} 5^{2\nu} B^2 s(t^4 + 10t^2s^2 + 5s^4).$$

Thus, if for brevity we put $2\mu - 1 = g$, $2\nu = h$, $B^2 = C$, everything reduces to showing that it is impossible to find two coprime numbers s and t , the first moreover even and divisible by 5, such that the product

$$2^g 5^h C s(t^4 + 10t^2s^2 + 5s^4) \quad (\beta)$$

is a fifth power.

Taking into account the nature of the prime divisors of C , one easily verifies that the factor $2^g 5^h C s$ and the trinomial factor are coprime. Thus, for the product (β) to be a fifth power, each of these factors would itself have to be one. The trinomial factor may be written in the form

$$(t^2 + 5s^2)^2 - 5(2s^2)^2.$$

Since the numbers $t^2 + 5s^2$ and $2s^2$ are evidently coprime, and moreover the first odd and the second even and divisible by 5, Theorem I is applicable here, and one may put

$$t^2 + 5s^2 = t'(t'^4 + 2 \cdot 5^2 t'^2 s'^2 + 5^3 s'^4),$$

$$2s^2 = 5s'(t'^4 + 10t'^2 s'^2 + 5s'^4).$$

The numbers s' and t' must be supposed coprime, and moreover the first even and the second odd and not divisible by 5. Since t' is not divisible by 5, it is evident from the second equation, whose first side contains the factor s^2 and is consequently divisible by 5^2 , that s' must be divisible by 5. Thus the numbers s' and t' are coprime, as are the numbers s and t , and the first, s' , is also even and divisible by 5, like s .

It is also easy to see that s' is smaller than s ; for from the last equation one immediately concludes that $5^2 s'^5 < 2s^2$, and hence that $s' < \sqrt[5]{\frac{2s^2}{25}}$.

We have seen above that $2^g 5^h C s$ had to be a fifth power. The number $2^{2g} 5^{2h} C^2 s^2$, its square, must therefore be a power of the same degree. Now, multiplying both sides of the last equation by $2^{2g-1} 5^{2h} C^2$, one obtains

$$2^{2g} 5^{2h} C^2 s^2 = 2^{2g-1} 5^{2h+1} C^2 s'(t'^4 + 10t'^2 s'^2 + 5s'^4),$$

or, what is the same thing, putting $2g - 1 = g'$, $2h + 1 = h'$, $C^2 = C'$ in the second side,

$$2^{2g} 5^{2h} C^2 s^2 = 2^{g'} 5^{h'} C' s'(t'^4 + 10t'^2 s'^2 + 5s'^4), \quad (\beta')$$

an equation whose two sides must be powers of the fifth degree.

We have therefore arrived at a product (β') similar to the product (β) , but in which the number s' is smaller than the number s of the product (β) ; and this product (β') would be a fifth power if

the product (β) were one. Treating the product (β') as we treated the product (β) , one would arrive at a third product (β'') , in which the number s'' would be smaller than s' , and this procedure could be continued as far as one wished. It is also easy to see that, however far one prolongs the series $s, s', s'', \dots$ and $t, t', t'', \dots$, one can never encounter a term equal to zero; for it is evident that if one supposed one of these terms to be zero, one would conclude, going backward, that $s = 0$, an evident case and one which, moreover, is excluded since the numbers s and t have been supposed coprime.

If, then, the product (β) could be a fifth power, one could obtain by the preceding analysis an indefinite sequence of positive integers, in which each term would be smaller than the preceding term, without any term being zero: this is a contradiction. One must conclude from this that the product (β) cannot be a power of the fifth degree.

Theorem II being thus established, we shall now give the means of deducing from it another, more general theorem. Consider the equation

$$x^5 \pm y^5 = 2^m Az^5,$$

in which we suppose x and y coprime, $m > 0$, and A subject to the same restrictions as in the statement of Theorem II. Let α, β, γ be positive numbers less than 5, and such that $x \equiv \alpha, \pm y \equiv \beta, z \equiv \gamma \pmod{5}$; and let H be a positive number < 25 such that $2^m A \equiv H \pmod{25}$. Since 5 is a prime number, one has

$$x^5 \equiv x, \quad y^5 \equiv y, \quad z^5 \equiv z \pmod{5}.$$

From this, taking into account the equation put above, one concludes

$$x \pm y \equiv 2^m Az \pmod{5},$$

and therefore

$$\alpha + \beta \equiv H\gamma \pmod{5}.$$

Since α is the remainder of x , one may put $x = \alpha + 5k$, where k is an integer; whence, raising both sides to the fifth power,

$$x^5 = \alpha^5 + 5\alpha^4(5k) + \frac{5 \cdot 4}{1 \cdot 2} \alpha^3(5k)^2 + \text{etc.},$$

and therefore

$$x^5 \equiv \alpha^5 \pmod{25}.$$

In the same manner one will find $\pm y^5 \equiv \beta^5, z^5 \equiv \gamma^5 \pmod{25}$; and since also $2^m A \equiv H \pmod{25}$, one finds, taking the cited equation into account,

$$\alpha^5 + \beta^5 \equiv H\gamma^5 \pmod{25}.$$

If now, substituting in this congruence successively for α and β all the combinations that can be formed with the positive numbers less than 5, and for γ the corresponding values, likewise less than 5, given by the formula

$$\alpha + \beta \equiv H\gamma \pmod{5},$$

one finds that the congruence $\alpha^5 + \beta^5 \equiv H\gamma^5 \pmod{25}$ can subsist only when γ is zero, then one will be assured that the equation

$$x^5 \pm y^5 = 2^m Az^5$$

cannot hold unless z is divisible by 5. In that case one may put $5z$ in place of z , which changes our equation into

$$x^5 \pm y^5 = 2^m 5^5 Az^5,$$

which evidently falls under Theorem II and is consequently impossible. Now this case occurs whenever the number H is one of the following eight numbers, 3, 4, 9, 12, 13, 16, 21, 22, as may be verified by a very simple calculation. Thus we have this new theorem:

Theorem III.

Let the numbers m and A be subject to the same restrictions as in the statement of Theorem II. If, on division by 25, the number $2^m A$ gives one of the following eight remainders, 3, 4, 9, 12, 13, 16, 21, 22, then it will be impossible to find two coprime numbers x and y such that

$$x^5 \pm y^5 = 2^m A z^5.$$

To give a very simple example, consider the following two equations:

$$x^5 \pm y^5 = 4z^5, \quad x^5 \pm y^5 = 16z^5.$$

Since in these equations one may, without loss of generality, suppose the numbers x and y coprime, it will be easy to see that they fall under Theorem II. Indeed, if one puts $A = 1$, and successively $m = 2$, $m = 4$, one has respectively

$$2^m A = 4, \quad 2^m A = 16.$$

Thus the two preceding equations are proved impossible.

Consider again the equation

$$x^5 \pm y^5 = z^5,$$

which is one of those that Fermat asserted to be impossible. By considerations similar to those which served us in establishing the preceding theorem, one may be assured that this equation cannot subsist unless one of the indeterminates x , y , z is divisible by 5. Let z be the indeterminate divisible by 5, for it is evident that one can arrange that any one of the indeterminates should stand alone in one side. On the other hand, if one supposes these indeterminates coprime, one of them will be even and the two others will be odd. If z were even, one could replace this indeterminate by $2 \cdot 5 \cdot z$, which would change the preceding equation into

$$x^5 \pm y^5 = 2^5 5^5 z^5,$$

which is impossible, since it evidently falls under Theorem II. It would therefore remain only to treat the case in which the indeterminate divisible by 5 is odd; but the method set out in this memoir appears insufficient for this case, and I do not see how one could complete the proof of the particular case of Fermat's theorem just mentioned.

Addition to the preceding memoir

This addition was presented to the Academy, and initialled by M. Fourier, the perpetual secretary, on 14 November 1825.

Since the preceding memoir was presented to the Academy, M. Legendre has published a second supplement to his Theory of Numbers, in which he proves the impossibility of the equation

$$x^5 \pm y^5 = z^5.$$

The case of the indeterminate divisible at the same time by 2 and by 5 is treated in that work as in the preceding memoir, and the author then proves the impossibility of the other case by a new analysis, though of the same kind as that used for the first case. The object of this addition is to establish two new theorems on indeterminate equations of the fifth degree, which contain, as a particular case, Fermat's theorem for fifth powers. To this end I rely on the results obtained above, and I use an analysis similar to that which M. Legendre used in the cited work, presenting it in such a manner as to show the great analogy which it has with the method set out in the preceding memoir.

Let the numbers P and Q , the first of which is supposed not divisible by 5, be both odd and have no common divisor. The number $P^2 - 5Q^2$ will be of the form $8k + 4$, and one may write

$$P^2 - 5Q^2 = 4L,$$

where L is an odd number not divisible by 5. If we multiply this equation side by side with the equation

$$3^2 - 5 \cdot 1^2 = 4,$$

we obtain

$$(3P \pm 5Q)^2 - 5(P \pm 3Q)^2 = 16L.$$

Since the numbers P and Q are both odd, it is evident that, by determining the sign suitably, the expressions

$$\frac{3P \pm 5Q}{4}, \quad \frac{P \pm 3Q}{4},$$

will both be integers. Setting accordingly

$$\frac{3P \pm 5Q}{4} = P', \quad \frac{\pm(P \pm 3Q)}{4} = Q',$$

the sign outside the parenthesis being chosen so as to give a positive value for Q' , the equation obtained above changes into $P'^2 - 5Q'^2 = L$, and one will easily verify that

$$P + Q\sqrt{5} = (P' \pm Q'\sqrt{5})(3 \pm \sqrt{5}),$$

the signs being suitably chosen, and that the numbers P' and Q' are coprime, and moreover one even and the other odd.

Now suppose that the number L is to be a fifth power. One satisfies this condition in the most general manner by putting

$$P' + Q'\sqrt{5} = (M \pm N\sqrt{5})^5(9 \pm 4\sqrt{5})^p.$$

Substituting this value in the last equation gives

$$P + Q\sqrt{5} = (M \pm N\sqrt{5})^5(9 \pm 4\sqrt{5})^p(3 \pm \sqrt{5}),$$

in which the signs are independent, as in the two preceding equations. One may put $p = 5k \pm r$, where k is a positive integer and r has one of the three values 0, 1, 2. The quantity $(9 \pm 4\sqrt{5})^p$ will then decompose into the two factors $(9 \pm 4\sqrt{5})^{5k}$ and $(9 \pm 4\sqrt{5})^{\pm r}$, of which the first may be omitted because it is absorbed into $(M \pm N\sqrt{5})^5$. If, moreover, we observe that in virtue of the equation

$$9 \pm 4\sqrt{5} = (9 \mp 4\sqrt{5})^{-1},$$

one may change in $(9 \pm 4\sqrt{5})^{\pm r}$ the signs of r and of the radical simultaneously, we may suppose the sign of r positive, and the equation given above becomes

$$P + Q\sqrt{5} = (M \pm N\sqrt{5})^5(9 \pm 4\sqrt{5})^r(3 \pm \sqrt{5}).$$

Since L must always be the fifth power of an odd number not divisible by 5, let us determine the necessary conditions for Q to be divisible by 5. Since the coefficient of $\sqrt{5}$ in the development of $(M \pm N\sqrt{5})^5$ is divisible by 5, while the rational part of this development is not, because M is not divisible by 5, one concludes, as in the proof of Theorem I, that in order for Q to be divisible by 5, the coefficient of $\sqrt{5}$ in the expanded value of $(9 \pm 4\sqrt{5})^r(3 \pm \sqrt{5})$ must be so divisible.

Now, substituting successively for r the three values 0, 1, 2, one finds that this occurs only in the case $r = 2$, the signs of the radicals in the two factors $(9 \pm 4\sqrt{5})^r$ and $(3 \pm \sqrt{5})$ being at the same time opposite. If one observes that

$$\frac{(3 \pm \sqrt{5})^3}{2^3} = 9 \pm 4\sqrt{5}, \quad 3 \mp \sqrt{5} = \frac{4}{3 \pm \sqrt{5}},$$

one will find that, in the case in question, the preceding product is equivalent to

$$\frac{(3 \pm \sqrt{5})^5}{2^4},$$

and substituting this value in the equation obtained above changes it into

$$P + Q\sqrt{5} = \frac{(M \pm N\sqrt{5})^5(3 \pm \sqrt{5})^5}{2^4}.$$

The numbers M and N being one even and the other odd, the numbers φ and ψ determined by the equation

$$\varphi + \psi\sqrt{5} = (M \pm N\sqrt{5})(3 \pm \sqrt{5})$$

will both be odd, and one will have

$$P + Q\sqrt{5} = \frac{(\varphi + \psi\sqrt{5})^5}{2^4}$$

and consequently

$$P = \varphi \frac{\varphi^4 + 2 \cdot 5^2 \varphi^2 \psi^2 + 5^3 \psi^4}{2^4}, \quad Q = 5\psi \frac{\varphi^4 + 10\varphi^2 \psi^2 + 5\psi^4}{2^4}.$$

For P and Q to be coprime, φ and ψ must have no common divisor and the first of these numbers must not be divisible by 5. Conversely, if the numbers φ and ψ , of which the first is supposed not to be divisible by 5, have no common divisor and are both odd, then the numbers P and Q will be integers and coprime. Indeed, the quarter of the quantity $\varphi^4 + 10\varphi^2 \psi^2 + 5\psi^4$ can be put in the form

$$\left(\frac{\varphi^2 + 5\psi^2}{2} \right)^2 - 5(\psi^2)^2,$$

and since this last expression is evidently the quadruple of an odd number, one sees that the value of Q is integral and odd. The same will be proved for the value of P , and one will easily verify that the numbers P and Q , which are both odd, have no common divisor. Thus we have this theorem:

Theorem IV.

Let the numbers P and Q be coprime and both odd, and let the latter be divisible by 5. I say that, in order to equate the binomial $P^2 - 5Q^2$ to the quadruple of a fifth power with all suitable generality, it will suffice to put

$$P + Q\sqrt{5} = \frac{(\varphi + \psi\sqrt{5})^5}{2^4},$$

the indeterminate numbers φ and ψ being coprime, both odd, and the first moreover not divisible by 5^5 .

Here now is the first of the new theorems announced at the beginning of this addition.

Theorem V.

Let the letter n denote a positive number other than 0 and 2, and let the number A be divisible neither by 2 nor by 5, nor by any prime number of either of the two forms $10k \pm 1$. Then it will be impossible to find two coprime numbers x and y such that

$$x^5 \pm y^5 = 5^n Az^5. \tag{\gamma}$$

The numbers x and y may either both be odd, or one even and the other odd. In the first case z will be divisible by 2, and one may put $2z$ in place of z , which changes equation (γ) into

$$x^5 \pm y^5 = 2^5 5^n Az^5,$$

which is impossible since it evidently falls under Theorem II. It remains therefore to prove the impossibility of the second case, in which the numbers x, y are supposed to be one even and the other odd. If we put

$$x \pm y = p, \quad x \mp y = q,$$

we shall have

$$2x = p + q, \quad \pm 2y = p - q,$$

and the numbers p and q will be coprime and both odd. Substituting the preceding values of $2x$ and $\pm 2y$ in equation (γ) , after multiplying both sides by 2^5 , gives

$$p(p^4 + 10p^2q^2 + 5q^4) = 2^4 5^n Az^5.$$

Since p must evidently be divisible by 5, put $p = 5r$, which gives

$$5^2 r(q^4 + 2 \cdot 5^2 q^2 r^2 + 5^3 r^4) = 2^4 5^n Az^5.$$

By hypothesis the number n is either equal to unity or greater than 2. If n is equal to unity, it will be necessary in the preceding equation to suppose z divisible by 5. In that case one may put $5z$ in place of z , or, what is the same thing, give 5 the exponent 6, whence one sees that in all cases one may suppose $n > 2$.

⁵This theorem, like Theorem I, has analogues for many other primes.

Putting the preceding equation in the form

$$r(q^4 + 2 \cdot 5^2 q^2 r^2 + 5^3 r^4) = 2^4 5^{n-2} A z^5,$$

and observing that $n > 2$ and that q is prime to $p = 5r$, one sees that r must be supposed divisible by 5.

Choose now a positive number ν such that $n + \nu - 2$ is divisible by 5, and a number B having no prime divisor other than the number A , and such that the product AB is a fifth power. Multiplying the last equation by $5^\nu B$, we obtain

$$5^\nu B r (q^4 + 2 \cdot 5^2 q^2 r^2 + 5^3 r^4) = 2^4 5^{n+\nu-2} A B z^5.$$

The trinomial factor can be put in the form $(q^2 + 5^2 r^2)^2 - 5(10r^2)^2$, and the numbers $q^2 + 5^2 r^2$ and $10r^2$ evidently have no common divisor other than 2. Thus all odd prime divisors of the trinomial factor will be of one of the forms $10k \pm 1$, and the trinomial factor will consequently be prime to B . It is evident that it is also prime to $5^\nu r$, and consequently to $5^\nu B r$.

Since the factor $5^\nu B r$ and the trinomial factor are coprime, and since the first of these factors is odd, it follows from the last equation, whose second side is the product of 2^4 and the fifth power of an odd number, that $5^\nu B r$ is a fifth power, and that the trinomial factor is a fifth power multiplied by 2^4 .

The quarter of the trinomial factor having to be the quadruple of a fifth power, and this quarter being puttable in the form

$$\left(\frac{q^2 + 5^2 r^2}{2} \right)^2 - 5(5r^2)^2,$$

where the numbers $\frac{q^2 + 5^2 r^2}{2}$ and $5r^2$ are evidently coprime, both odd, and the latter moreover divisible by 5, it will suffice, by the theorem established at the beginning of this addition, in order to equate the quarter of the trinomial factor to the quadruple of a fifth power, to put the two equations

$$\begin{aligned} \frac{q^2 + 5^2 r^2}{2} &= t \frac{t^4 + 2 \cdot 5^2 t^2 s^2 + 5^3 s^4}{2^4}, \\ 5r^2 &= 5s \frac{t^4 + 10t^2 s^2 + 5s^4}{2^4}, \end{aligned}$$

the indeterminate numbers t and s being supposed coprime, both odd, and the first moreover not divisible by 5. Since t is not divisible by 5, while r is so divisible, as we have seen above, it follows from the last of the preceding equations that s must be divisible by 5.

We have seen above that $5^\nu B r$ had to be a fifth power. The number $5^{2\nu} B^2 r^2$, the square of the preceding number, must therefore be a power of the same degree, and even of the tenth degree. Now, multiplying both sides of the last equation by $2^4 \cdot 5^{2\nu-1} B^2$, one obtains

$$2^4 5^{2\nu} B^2 r^2 = 5^{2\nu} B^2 s (t^4 + 10t^2 s^2 + 5s^4).$$

Thus, if for brevity we put $2\nu = h$, $B^2 = C$, everything is reduced to showing that it is impossible to find two coprime numbers t and s , both odd and the latter s moreover divisible by 5, such that the product

$$5^h C s (t^4 + 10t^2 s^2 + 5s^4) \tag{\delta}$$

is a fifth power multiplied by 2^4 .

It is easy to see that the product (δ) cannot be the product of 2^4 and a fifth power unless 5^hCs is a fifth power and the trinomial factor is a fifth power multiplied by 2^4 .

The quarter of the trinomial factor having to be the quadruple of a fifth power, and this quarter being puttable in the form

$$\left(\frac{t^2 + 5s^2}{2}\right)^2 - 5(s^2)^2,$$

where the numbers $\frac{t^2 + 5s^2}{2}$ and s^2 are evidently coprime, both odd, and the latter moreover divisible by 5, it will suffice, in order to equate the trinomial factor divided by 4 to the quadruple of a fifth power, to put the equations

$$\begin{aligned}\frac{t^2 + 5s^2}{2} &= t' \frac{t'^4 + 2 \cdot 5^2 t'^2 s'^2 + 5^3 s'^4}{2^4}, \\ s^2 &= 5s' \frac{t'^4 + 10t'^2 s'^2 + 5s'^4}{2^4},\end{aligned}$$

the numbers t' and s' being supposed coprime, both odd, and the first t' moreover not divisible by 5. Since s is divisible by 5 and t' is no longer so, it follows from the last equation that s' is also divisible by 5. One concludes again from the last equation that $\frac{25}{16}s'^5 < s^2$ and hence that s' is much smaller than s .

The number 5^hCs being a fifth power, $5^{2h}C^2s^2$, the square of the preceding number, must be a power of the same degree, and even of the tenth degree. Now, multiplying both sides of the last equation by $2^4 5^{2h}C^2$, one obtains

$$2^4 5^{2h} C^2 s^2 = 5^{2h+1} C^2 s' (t'^4 + 10t'^2 s'^2 + 5s'^4),$$

or what is the same thing, putting $2h + 1 = h'$, $C^2 = C'$ in the second side,

$$2^4 5^{2h} C^2 s^2 = 5^{h'} C' s' (t'^4 + 10t'^2 s'^2 + 5s'^4), \quad (\delta')$$

an equation whose two sides must be fifth powers multiplied by 2^4 .

The product (δ') being perfectly similar to the product (δ) , and the number s' being much smaller than s , one will conclude, as in the proof of Theorem II of the preceding memoir, that the product (δ) cannot be equal to a fifth power multiplied by 2^4 , and that consequently equation (γ) cannot hold.

Fermat's theorem for the case of fifth powers is included as a particular case in the theorem which we have just established. Indeed, since the equation $x^5 \pm y^5 = z^5$ cannot hold unless one of the indeterminates, z for example, is divisible by 5, we may put $5z$ in place of z ; this gives $x^5 \pm y^5 = 5^5 z^5$, an impossible equation, since it falls under the last theorem.

A reasoning entirely similar to that by which we established Theorem III from Theorem II can serve to deduce from the theorem just proved a new theorem which may be stated as follows:

Theorem VI.

Let the number A be subject to the same restrictions as in the statement of Theorem V, and suppose that, when divided by 25, this number gives one of the following eight remainders, 3, 4, 9, 12, 13, 16, 21, 22. Then it will be impossible to find two coprime numbers x and y such that

$$x^5 \pm y^5 = Az^5.$$

END.

Memoir on the impossibility of some indeterminate equations of the fifth degree

by

Mr. Lejeune Dirichlet,
Professor of mathematics

Read by the author to the Royal Academy of Sciences (Institut de France), 11 July 1825.

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Memoir on the impossibility of certain indeterminate equations of the fifth degree

Read to the Royal Academy of Sciences (Institut de France) on 11 July 1825, by the author.

According to the report of Messrs. Lacroix and Legendre, this memoir was approved and is to be printed in the collection of memoirs by foreign scholars¹.

It is known that the theory of indeterminate equations of degrees higher than the second is still very little advanced. It is true that there are infinitely many equations of all degrees whose impossibility can be shown by proving that, whatever values are attributed to the unknowns, the two sides of the proposed equation can never give the same remainder when divided by a certain number, or modulus. But when an equation cannot be treated by this means, it becomes difficult to prove that it is impossible, and up to the present this has been achieved only for a very small number of equations. All these equations are very special, and of such a form that, when one seeks to solve them, one is naturally led to one or more quadratic formulae that must be made equal to perfect powers. One then satisfies this condition in the most general way by expressing the unknowns by means of others, of which the former become integral functions; and it turns out, at least in all cases in which the method in question succeeds, that the new unknowns, or other quantities depending on them in a very simple way, likewise satisfy an equation similar to the proposed equation. Since the new unknowns are at the same time smaller than the primitive unknowns, the impossibility of the proposed equation is thereby established: for it is evident that, if it were possible, one would have a means of obtaining a decreasing and indefinite sequence of integers, which implies a contradiction. It is in this way that Fermat and Euler proved the impossibility of several equations of the third and fourth degrees.

When one tries to apply similar considerations to certain equations of the fifth degree and of a form analogous to those of the equations treated by Fermat and Euler, one is immediately stopped. The quadratic formula at which one arrives, and which has to be made equal to a fifth power, admits several different solutions; and among these solutions there is only one that leads to an equation similar to the proposed equation. Reflecting on this difficulty, I recognized that it could be removed very simply by subjecting to several conditions the determined number that enters into the equation. It follows from a theorem stated in the preliminaries that, when these conditions are fulfilled, the different solutions of which the quadratic formula is susceptible in general must be rejected, with the exception of a single one, which is precisely that from which one deduces numbers satisfying an equation similar to the proposed equation. One thus succeeds in establishing the impossibility of a fairly extensive class of indeterminate equations of the fifth degree. The first member of these equations is the sum or the difference of two fifth powers, and the second member is the product of a fifth power and a determined number subject to various conditions. By assigning to this number particular values compatible with these conditions, one can obtain as many particular theorems as one wishes. This generality of our theorems is all the more remarkable because the analogous equations of the third and fourth degrees whose impossibility has been proved up to the present are only finite in number, and indeed very few.

Theorem I.

Let l be an odd prime number not dividing the number a , and suppose that

$$\delta^2 - a\varepsilon^2 = l. \tag{1}$$

¹This Memoir has not yet been published up to the present. Editor's note.

It is known that the equation

$$d^2 - ae^2 = l^n \quad (2)$$

is satisfied by the numbers d and e given by the formula

$$(\delta + \varepsilon\sqrt{a})^n = d + e\sqrt{a}, \quad (3)$$

when the rational parts and the coefficients of $\sqrt{a}$ are equated. I further say that the numbers d and e thus obtained are relatively prime².

It is evident from equation (2) that if the numbers d and e had a common divisor, it could only be the number l or a power of that number. It will therefore suffice to show that d is not divisible by l . Formula (3) gives this value of d :

$$d = \delta^n + \frac{n(n-1)}{1 \cdot 2} a \delta^{n-2} \varepsilon^2 + \frac{n(n-1)(n-2)(n-3)}{1 \cdot 2 \cdot 3 \cdot 4} a^2 \delta^{n-4} \varepsilon^4 + \text{etc.}$$

On the other hand, from equation (1), using the sign employed by Mr. Gauss, one concludes

$$\delta^2 \equiv a\varepsilon^2, \quad \delta^4 \equiv a^2\varepsilon^4, \quad \delta^6 \equiv a^3\varepsilon^6, \quad \dots \pmod{l},$$

and, multiplying respectively by $\delta^{n-2}, \delta^{n-4}, \dots$,

$$\delta^n \equiv a\delta^{n-2}\varepsilon^2, \quad \delta^n \equiv a^2\delta^{n-4}\varepsilon^4, \quad \dots \pmod{l}.$$

Combining these congruences with the equation that gives d , one obtains

$$d \equiv \delta^n \left(1 + \frac{n(n-1)}{1 \cdot 2} + \frac{n(n-1)(n-2)(n-3)}{1 \cdot 2 \cdot 3 \cdot 4} + \text{etc.} \right),$$

or, what is the same thing, since the quantity in brackets is the expansion of $\frac{1}{2}[(1+1)^n + (1-1)^n]$ and consequently equal to 2^{n-1} ,

$$d \equiv 2^{n-1} \delta^n \pmod{l}.$$

It is now easy to see that d is not divisible by l , for for that to happen δ would have to be divisible by l ; but inspection of equation (1) alone shows that this is impossible, δ and ε being evidently relatively prime and a not divisible by l .

Theorem II.

Let the letter l designate an odd prime not dividing a . If one supposes that

$$d^2 - ae^2 = l^n, \quad (4)$$

$$d'^2 - ae'^2 = l^n, \quad (5)$$

the numbers d and e , d' and e' being relatively prime, I say that one can find two numbers t and u satisfying

$$t^2 - au^2 = 1 \quad (6)$$

and also such that

$$(d' \pm e'\sqrt{a})(t \pm u\sqrt{a}) = d + e\sqrt{a}, \quad (7)$$

the signs being suitably chosen and the rational parts and the coefficients of $\sqrt{a}$ equated separately.

²If the number l , instead of being prime, were any odd number and the numbers δ and $a\varepsilon$ were relatively prime, the numbers d and ae would likewise be relatively prime, as is easily verified by a reasoning perfectly similar to that used in the text.

Equations (4) and (5) give immediately

$$d^2 \equiv ae^2, \quad d'^2 \equiv ae'^2 \pmod{l^n};$$

therefore, multiplying side by side,

$$d^2 d'^2 \equiv a^2 e^2 e'^2 \pmod{l^n},$$

and by transposing,

$$d^2 d'^2 - a^2 e^2 e'^2 \equiv (dd' + aee')(dd' - aee') \equiv 0 \pmod{l^n}.$$

This congruence shows that one of the numbers $dd' + aee'$, $dd' - aee'$ is divisible by l^n , or that they are both divisible by l^3 .

But it is easy to see that this last case is impossible. Indeed, if these numbers were both divisible by l , their sum $2dd'$ would also be divisible by l ; hence in that case one of the numbers d , d' would have to be divisible by l . But one easily verifies that this cannot be, taking account of the hypotheses made in the statement of the theorem. The expression

$$\frac{dd' \pm aee'}{l^n}$$

with the proper sign will therefore be an integer. For greater simplicity we shall put $i = \pm 1$, i being chosen so as to make the preceding expression integral.

If one multiplies equations (4) and (5) side by side, one obtains

$$(dd' \pm aee')^2 - a(d'e \pm de')^2 = l^{2n},$$

in which one may take either the upper signs or the lower signs at will. Therefore one also has

$$(dd' + iaee')^2 - a(d'e + ide')^2 = l^{2n}.$$

Since the number $dd' + iaee'$ is divisible by l^n , and a is not divisible by l , one sees from the preceding equation that $d'e + ide'$ is likewise divisible by l^n . This being so, I say that one has

$$t = \frac{dd' + iaee'}{l^n}, \quad u = \frac{i'(d'e + ide')}{l^n},$$

where i' is 1 or -1 according as the quantity in parentheses is positive or negative. Indeed, dividing the two numbers in the equation given above by l^n , one has

$$\left(\frac{dd' + iaee'}{l^n}\right)^2 - a\left(\frac{d'e + ide'}{l^n}\right)^2 = 1,$$

or, what is the same thing, $t^2 - au^2 = 1$, and by substitution one easily verifies that the preceding values of t and u also satisfy equation (7), when the signs are determined in the following way:

$$(d' - ie'\sqrt{a})(t + i'u\sqrt{a}) = d + e\sqrt{a}.$$

Remark. Since it is evident that one can change simultaneously the signs of e' , u , e in the equation

$$(d' \pm e'\sqrt{a})(t \pm u\sqrt{a}) = d + e\sqrt{a},$$

one sees that one may put

$$(d' \pm e'\sqrt{a})(t \pm u\sqrt{a}) = d \pm e\sqrt{a},$$

the sign of e being arbitrary and the signs of e' and u being suitably chosen.

³In the case $n = 1$, the first hypothesis is included in the second and therefore necessarily holds; but the same reasoning would still prove the impossibility of the second.

Theorem III.

Let l denote an odd prime not dividing a , and let k be an odd number that has no common divisor with a and is not divisible by l . If we suppose that the two equations $D^2 - aE^2 = l^n k$, $d^2 - ae^2 = l^n$ hold, and that the numbers D and E , d and e are relatively prime, I say that one can find two numbers D' and E' , relatively prime, satisfying the equation $D'^2 - aE'^2 = k$, and also such that

$$(D' \pm E'\sqrt{a})(d \pm e\sqrt{a}) = D + E\sqrt{a},$$

the signs being suitably chosen and the rational parts and the coefficients of $\sqrt{a}$ being equated separately.

The proof of this theorem is so similar to that of Theorem II that we shall refrain from developing it here.

There is here a similar remark to be made, and one sees very easily that one may put

$$(D' \pm E'\sqrt{a})(d \pm e\sqrt{a}) = D \pm E\sqrt{a},$$

the sign of E being arbitrary and the signs of E' and e being suitably chosen.

The theorems that we have just established, and which may be useful on several occasions, will now serve us to establish a proposition concerning the way to make the binomial $P^2 - 5Q^2$, in which P and Q are indeterminate numbers subject to the restriction of being relatively prime, equal to a fifth power. This proposition consists in the following: to make the binomial $P^2 - 5Q^2$ equal, in the most general manner, to an odd fifth power not divisible by 5, one need only put

$$P + Q\sqrt{5} = (M \pm N\sqrt{5})^5(t \pm u\sqrt{5}),$$

where M and N are new unknowns subject to the sole restriction of being relatively prime, one even, the other odd, and the first moreover not divisible by 5; and the letters t and u designate the general solution of the equation $t^2 - 5u^2 = 1$. That is to say, whenever P and Q are relatively prime and the binomial $P^2 - 5Q^2$ is an odd fifth power not divisible by 5, there exist numbers M and N , relatively prime, such that

$$P + Q\sqrt{5} = (M \pm N\sqrt{5})^5(t \pm u\sqrt{5}),$$

with t and u satisfying $t^2 - 5u^2 = 1$.

To verify the truth of this proposition, put $P^2 - 5Q^2 = L$, where L denotes an odd fifth power not divisible by 5, and let us see what follows concerning the nature of the numbers P and Q . Let l denote any one of the prime divisors of L , and let l^{5n} be the highest power of l that divides L , so that, putting $L = l^{5n}L'$, L' is prime to l . It follows from a known theorem that the number l dividing $P^2 - 5Q^2$ will itself be of the form $\delta^2 - 5\varepsilon^2$. If we now put

$$(\delta + \varepsilon\sqrt{5})^n = d + e\sqrt{5}, \quad (\delta + \varepsilon\sqrt{5})^{5n} = D + E\sqrt{5},$$

then d and e , D and E are relatively prime by Theorem I, and one has

$$(d + e\sqrt{5})^5 = D + E\sqrt{5}.$$

If one then applies Theorem III to the equations

$$D^2 - 5E^2 = l^{5n}, \quad P^2 - 5Q^2 = l^{5n}L',$$

which follow immediately from what precedes, one sees that there exist numbers P' and Q' , relatively prime, such that

$$P'^2 - 5Q'^2 = L', \quad P + Q\sqrt{5} = (D \pm E\sqrt{5})(P' \pm Q'\sqrt{5}),$$

or, what amounts to the same thing, replacing $D + E\sqrt{5}$ by $(d + e\sqrt{5})^5$,

$$P + Q\sqrt{5} = (d \pm e\sqrt{5})^5(P' + Q'\sqrt{5}).$$

The equation $P'^2 - 5Q'^2 = L'$ at which we have just arrived is entirely analogous to $P^2 - 5Q^2 = L$, for the number L' obtained by dividing L by l^{5n} is a fifth power, like L . Suppose for a moment that the proposition we seek to prove is true for the equation $P'^2 - 5Q'^2 = L'$, and let us see how one could infer that it also holds for the binomial $P^2 - 5Q^2$. Under the supposition just made, there exist numbers M' and N' such that

$$P' + Q'\sqrt{5} = (M' \pm N'\sqrt{5})^5(t \pm u\sqrt{5}).$$

Putting this value of $P' + Q'\sqrt{5}$ into the equation obtained above, whose first member contains $P + Q\sqrt{5}$, gives

$$P + Q\sqrt{5} = (M' \pm N'\sqrt{5})^5(d \pm e\sqrt{5})^5(t \pm u\sqrt{5}),$$

where the signs depend on those found in the two equations whose combination has produced it. If we now put

$$(M' \pm N'\sqrt{5})(d \pm e\sqrt{5}) = M \pm N\sqrt{5},$$

the sign of N being $+$ or $-$ according as the coefficient of $\sqrt{5}$ in the developed value of the first member is positive or negative, the preceding equation becomes

$$P + Q\sqrt{5} = (M \pm N\sqrt{5})^5(t \pm u\sqrt{5}),$$

which conforms to the statement of the proposition under consideration.

Having thus shown that the proposition in question is true for the binomial $P^2 - 5Q^2$ equal to the fifth power L , if it is supposed to hold for the binomial $P'^2 - 5Q'^2$ equal to the fifth power L' , which has one prime divisor less than L , it remains, to make the proof complete, only to prove the truth of our proposition in the case where $P^2 - 5Q^2$ is a fifth power with only one prime divisor l . This is very easy to do using Theorem II. Indeed, in the case just stated one has $P^2 - 5Q^2 = l^{5n}$; on the other hand, since the number l can be put under the form $\delta^2 - 5\varepsilon^2$, if one sets

$$(\delta + \varepsilon\sqrt{5})^n = M + N\sqrt{5}, \quad (\delta + \varepsilon\sqrt{5})^{5n} = D + E\sqrt{5},$$

one also has

$$D + E\sqrt{5} = (M + N\sqrt{5})^5, \quad D^2 - 5E^2 = l^{5n},$$

with D and E relatively prime by Theorem I. This being so, application of Theorem II to the equations

$$P^2 - 5Q^2 = l^{5n}, \quad D^2 - 5E^2 = l^{5n}$$

immediately gives the relation

$$P + Q\sqrt{5} = (D \pm E\sqrt{5})(t \pm u\sqrt{5}),$$

where t and u satisfy $t^2 - 5u^2 = 1$ and the signs are suitably chosen; or, what amounts to the same thing, replacing $D \pm E\sqrt{5}$ by $(M \pm N\sqrt{5})^5$,

$$P + Q\sqrt{5} = (M \pm N\sqrt{5})^5(t \pm u\sqrt{5}),$$

a result conforming to the statement of our proposition. It is thus proved that whenever $P^2 - 5Q^2$ is an odd fifth power, there exist numbers M and N satisfying the preceding formula. As for the converse of this proposition, namely that by assigning in the preceding formula to M and N values subject only to the restrictions already stated several times, one obtains numbers P and Q relatively prime and such that $P^2 - 5Q^2$ is a fifth power, its proof is so simple that it is useless to pause over it. By means of what precedes it will be easy to establish the theorem we are about to state and which will serve as the basis for the propositions that form the principal object of this memoir.

Theorem IV.

The numbers P and Q being relatively prime, one even, the other odd, and the latter moreover divisible by 5, I say that to make the binomial $P^2 - 5Q^2$ equal, in the most general way, to a fifth power, it suffices to put

$$P + Q\sqrt{5} = (\varphi + \psi\sqrt{5})^5.$$

The unknowns φ and ψ are relatively prime, one even, the other odd, and the first moreover not divisible by 5⁴.

To make $P^2 - 5Q^2$ equal to a fifth power, we shall put, by what has just been said,

$$P + Q\sqrt{5} = (M \pm N\sqrt{5})^5(t \pm u\sqrt{5}).$$

The number N not being divisible by 5, if for a moment we put

$$(M \pm N\sqrt{5})^5 = M' + N'\sqrt{5},$$

it will be easy to see that N' is divisible by 5 and that M' is not. Substituting the preceding expression into the value of $P + Q\sqrt{5}$, one obtains

$$P + Q\sqrt{5} = (M' \pm N'\sqrt{5})(t \pm u\sqrt{5}),$$

whence

$$Q = \pm M'u \pm N't.$$

Since N' is divisible by 5 and M' is not, it is evident that Q can be divisible by 5 only if u is. The smallest values satisfying

$$t^2 - 5u^2 = 1$$

are

$$t = 9, \quad u = 4.$$

The general values are therefore given by

$$t + u\sqrt{5} = (9 + 4\sqrt{5})^p,$$

⁴It is perhaps not useless to point out that there are analogous theorems for many other prime numbers, and that to establish them one can use the same considerations that we use here.

where p is any positive integer⁵. From this one obtains

$$u = \frac{p}{1} 9^{p-1} \cdot 4 + \frac{p(p-1)(p-2)}{1 \cdot 2 \cdot 3} 9^{p-3} \cdot 4^3 \cdot 5 + \text{etc.}$$

All terms, beginning with the second, being divisible by 5 for every p , one sees that, for u to be divisible by 5, the first term, and consequently also p , must be divisible by 5. If then we put $p = 5p'$, with p' an integer, and substitute the value of $t + u\sqrt{5}$ in that of $P + Q\sqrt{5}$, we get

$$P + Q\sqrt{5} = (M \pm N\sqrt{5})^5 (9 \pm 4\sqrt{5})^{5p'},$$

a result which, by introducing new unknowns φ and ψ such that

$$(M \pm N\sqrt{5})(9 \pm 4\sqrt{5})^{p'} = \varphi + \psi\sqrt{5},$$

becomes

$$P + Q\sqrt{5} = (\varphi + \psi\sqrt{5})^5.$$

The form of the solution given by the theorem now being justified, it remains only to determine the nature of the unknowns. Since

$$P^2 - 5Q^2 = (\varphi^2 - 5\psi^2)^5,$$

and the numbers P and Q are respectively divisible by φ and ψ , one easily sees that the unknowns φ and ψ must be supposed relatively prime, one even, the other odd, and the first moreover not divisible by 5. One may even add that φ or ψ will be odd according as P or Q is odd. Conversely, if the unknowns φ and ψ satisfy the preceding conditions, the numbers P and Q determined by

$$P + Q\sqrt{5} = (\varphi + \psi\sqrt{5})^5$$

will be relatively prime, as is easily verified. These preliminaries established, we can consider the theorems that form the principal object of this memoir. The first can be stated as follows.

Theorem V.

Let the numbers m and n be positive, greater than zero, the second moreover different from 2, and let the number A be divisible neither by 2 nor by 5, nor by any prime number of the form $10k + 1$. It will be impossible to find two relatively prime numbers x and y such that

$$x^5 \pm y^5 = 2^m 5^n A z^5. \tag{\alpha}$$

Suppose, contrary to the theorem, that the equation is possible. Since the second member is even (as m has been supposed > 0), the relatively prime numbers x and y must both be odd. If we set $x \pm y = 2p$, $x \mp y = 2q$, and consequently

$$x = p + q, \quad \pm y = p - q,$$

then p and q are integers, relatively prime, and one is even and the other odd. Substituting the preceding values of x and $\pm y$ into equation (α) transforms it into

$$2p(p^4 + 10p^2q^2 + 5q^4) = 2^m 5^n A z^5.$$

⁵See the Additions to Euler's Algebra (art. 75).

The first member can be equal to the second, which is divisible by 5, only if p is divisible by 5. Putting therefore $p = 5r$, we get

$$2 \cdot 5^2 r (q^4 + 2 \cdot 5^2 q^2 r^2 + 5^3 r^4) = 2^m 5^n A z^5.$$

By hypothesis, n is equal to unity or greater than 2. If n is equal to unity, then in the preceding equation z must be supposed divisible by 5. In that case one can put $5z$ in place of z , or, what is the same thing, give 5 the exponent 6, whence one sees that in all cases one may suppose $n > 2$. If now one puts the preceding equation in the form

$$r(q^4 + 2 \cdot 5^2 q^2 r^2 + 5^3 r^4) = 2^{m-1} 5^{n-2} A z^5,$$

and recalls that the numbers q and $p = 5r$ are relatively prime, and moreover one even and the other odd, it is easy to see that the trinomial factor is odd, not divisible by 5, and prime to r ; therefore r must be divisible by 5, since $n > 2$. Now choose two positive numbers μ and ν such that⁶ $m + \mu - 1$ and $n + \nu - 2$ are divisible by 5, and a number B having no prime divisors other than those of A and such that the product AB is a fifth power. If we multiply the preceding equation by $2^\mu 5^\nu B$, we get

$$2^\mu 5^\nu B r (q^4 + 2 \cdot 5^2 q^2 r^2 + 5^3 r^4) = 2^{m+\mu-1} 5^{n+\nu-2} A B z^5.$$

The second member being a fifth power, the first member is also a fifth power. I now say that the two factors into which this first member can be decomposed, the factor $2^\mu 5^\nu B r$ and the trinomial factor, are relatively prime. Indeed, we have already seen that the trinomial factor is prime to $2^\mu 5^\nu r$, and, on the other hand, it follows from Euler's known theorems on the linear form of the prime divisors of the formula $x^n + y^n$ (*Theory of Numbers*, no. 156) that the trinomial factor, in which q and r have no common divisor, is divisible only by prime numbers of the form $10k + 1$; this form, by the hypotheses in the statement of the theorem, is not suitable for any divisor of A , and consequently not for any divisor of B , since B has no prime divisors other than those of A . Hence the number $2^\mu 5^\nu B r$ and the trinomial factor must both be fifth powers.

The trinomial factor can be written as

$$(q^2 + 5^2 r^2)^2 - 5(10r^2)^2,$$

and the numbers $q^2 + 5^2 r^2$, $10r^2$ are relatively prime, the first odd, the second even and divisible by 5. Therefore, by Theorem I, in order to make the trinomial factor equal, with the proper generality, to a fifth power, it will suffice to put

$$q^2 + 5^2 r^2 = t(t^4 + 2 \cdot 5^2 t^2 s^2 + 5^3 s^4),$$

$$10r^2 = 5s(t^4 + 10t^2 s^2 + 5s^4).$$

The numbers s and t must be supposed relatively prime, the first even, the second odd and not divisible by 5. It follows that s must be divisible by 5: indeed, since t is not divisible by 5, the second member of the second equation can be divisible by 5^2 only if s is divisible by 5; but the first member, which has r^2 as a factor, is divisible by 5^2 ; hence s is divisible by 5.

We have seen above that $2^\mu 5^\nu B r$ had to be a fifth power. The number $2^{2\mu} 5^{2\nu} B^2 r^2$, the square of the preceding number, must therefore be a power of the same degree, indeed of the tenth degree. Now, multiplying both sides of the last equation by $2^{2\mu-1} 5^{2\nu-1} B^2$, one obtains

$$2^{2\mu} 5^{2\nu} B^2 r^2 = 2^{2\mu-1} 5^{2\nu} B^2 s(t^4 + 10t^2 s^2 + 5s^4).$$

⁶If $m - 1$ were divisible by 5, one would choose for μ another value than zero in order to avoid negative exponents in what follows.

Thus, writing for abbreviation $2\mu - 1 = g$, $2\nu = h$, $B^2 = C$, everything reduces to showing that it is impossible to find two relatively prime numbers s and t , the first moreover even and divisible by 5, such that the product

$$(\beta) \quad 2^g 5^h C s (t^4 + 10t^2 s^2 + 5s^4)$$

is a fifth power.

Taking account of the nature of the prime divisors of C , one verifies easily that the factor $2^g 5^h C s$ and the trinomial factor are relatively prime. For the product (β) to be a fifth power, each of these factors would have to be one. The trinomial factor can be written as

$$(t^2 + 5s^2)^2 - 5(2s^2)^2.$$

Since the numbers $t^2 + 5s^2$ and $2s^2$ are evidently relatively prime, and the first is odd while the second is even and divisible by 5, Theorem I is applicable here, and one may put

$$t^2 + 5s^2 = t'(t'^4 + 2 \cdot 5^2 t'^2 s'^2 + 5^3 s'^4),$$

$$2s^2 = 5s'(t'^4 + 10t'^2 s'^2 + 5s'^4).$$

The numbers s' and t' must be supposed relatively prime, the first even, the second odd and not divisible by 5. Since t' is not divisible by 5, it is evident from the second equation, whose first member contains the factor s^2 and is consequently divisible by 5^2 , that s' must be divisible by 5. Thus the numbers s' and t' are relatively prime, just as the numbers s and t are, and the first, s' , is moreover even and divisible by 5 like s .

It is still easy to see that s' is smaller than s ; for from the last equation one immediately concludes $5^2 s'^5 < 2s^2$, and hence $s' < \sqrt[5]{\frac{2s^2}{25}} < s$. We saw above that $2^g 5^h C s$ had to be a fifth power. The number $2^{2g} 5^{2h} C^2 s^2$, the square of the preceding one, must therefore be a power of the same degree. Multiplying both sides of the last equation by $2^{2g-1} 5^{2h} C^2$, one gets

$$2^{2g} 5^{2h} C^2 s^2 = 2^{2g-1} 5^{2h+1} C^2 s' (t'^4 + 10t'^2 s'^2 + 5s'^4),$$

or, what is the same thing, putting $2g - 1 = g'$, $2h + 1 = h'$, $C^2 = C'$ in the second member,

$$(\beta') \quad 2^{2g} 5^{2h} C^2 s^2 = 2^{g'} 5^{h'} C' s' (t'^4 + 10t'^2 s'^2 + 5s'^4),$$

an equation whose two members must be fifth powers.

We have therefore reached a product (β') similar to the product (β) , but in which the number s' is smaller than the number s of the product (β) ; and this product (β') would be a fifth power if the product (β) were one. Treating the product (β') as we treated the product (β) , one would arrive at a third product (β'') , in which the number s'' would be smaller than s' , and one could continue this procedure as far as desired. It is also easy to see that, however far one prolongs the series $s, s', s'', \dots; t, t', t'', \dots$, one can never meet a term equal to zero; for it is evident that, if one of these terms were supposed zero, on going back one would conclude that $s = 0$, an obvious case and, moreover, one excluded because the numbers s and t were supposed relatively prime.

If therefore the product (β) could be a fifth power, the preceding analysis would give an indefinite sequence of positive integers in which each term is smaller than the preceding term, without any term being zero; this implies a contradiction. One must conclude from this that the product (β) cannot be a power of the fifth degree.

Theorem V thus established, we shall now give a way to deduce from it another, more general theorem. Consider the equation

$$x^5 \pm y^5 = 2^m Az^5,$$

in which we suppose x and y relatively prime, $m > 0$, and A subject to the same restrictions as in the statement of Theorem V. Let α, β, γ be positive numbers less than 5, such that $x = \alpha, \pm y = \beta, z = \gamma \pmod{5}$, and let H be a positive number < 25 such that $2^m A = H \pmod{25}$. Since 5 is a prime number, one has

$$x^5 \equiv \alpha, \quad \pm y^5 \equiv \beta, \quad z^5 \equiv \gamma \pmod{5}.$$

From this, taking account of the equation posed above, one concludes

$$\alpha + \beta \equiv H\gamma \pmod{5}.$$

Since α is the remainder of x , one may put $x = \alpha + 5k$, where k is an integer. Raising both sides to the fifth power gives

$$x^5 = \alpha^5 + 5\alpha^4(5k) + \frac{5 \cdot 4}{1 \cdot 2}\alpha^3(5k)^2 + \text{etc.},$$

and hence

$$x^5 \equiv \alpha^5 \pmod{25}.$$

In the same way one finds $\pm y^5 \equiv \beta^5, z^5 \equiv \gamma^5 \pmod{25}$; and since also $2^m A \equiv H \pmod{25}$, one obtains, taking account of the cited equation,

$$\alpha^5 + \beta^5 \equiv H\gamma^5 \pmod{25}.$$

If now, substituting successively in this congruence for α, β all combinations that can be formed with the positive numbers less than 5, and for γ the corresponding values, also less than 5, given by

$$\alpha + \beta \equiv H\gamma \pmod{5},$$

one finds that the congruence $\alpha^5 + \beta^5 \equiv H\gamma^5 \pmod{25}$ can hold only when γ is zero, then one is certain that the equation

$$x^5 \pm y^5 = 2^m Az^5$$

cannot hold unless z is divisible by 5. In that case one may put $5z$ in place of z , which changes our equation into

$$x^5 \pm y^5 = 2^m 5^5 Az^5,$$

which evidently falls under Theorem II and is therefore impossible. Now this case occurs whenever the number H is one of the following eight numbers, 3, 4, 9, 12, 13, 16, 21, 22, as one may verify by a very simple calculation. Thus we have this new theorem.

Theorem VI.

Let the numbers m and A be subject to the same restrictions as in the statement of Theorem II. If the number $2^m A$, when divided by 25, gives one of the eight remainders 3, 4, 9, 12, 13, 16, 21, 22, it will be impossible to find two relatively prime numbers x and y such that $x^5 \pm y^5 = 2^m Az^5$.

As a very simple example, consider the two equations

$$x^5 \pm y^5 = 4z^5, \quad x^5 \pm y^5 = 16z^5.$$

Since in these equations one may suppose, without loss of generality, that x and y are relatively prime, it is easy to see that they fall under Theorem II. Indeed, if one takes $A = 1$ and successively $m = 2$, $m = 4$, one obtains respectively

$$2^m A = 4, \quad 2^m A = 16.$$

It is therefore proved that the two preceding equations are impossible.

Consider again the equation

$$x^5 \pm y^5 = z^5,$$

which is one of those that Fermat asserted to be impossible. By considerations similar to those used to establish the preceding theorem, one can make sure that this equation cannot hold unless one of the unknowns x , y , z is divisible by 5. Let z be the unknown divisible by 5, for evidently one can arrange that any one of the unknowns stands alone in one member. On the other hand, if these unknowns are supposed pairwise prime, one of them will be even and the two others odd. If z were even, one could replace it by $2 \cdot 5z$, which would transform the preceding equation into

$$x^5 \pm y^5 = 2^5 5^5 z^5,$$

which is impossible, since it plainly falls under Theorem II. It would remain only to treat the case in which the unknown divisible by 5 is odd; but the method set out in this memoir appears insufficient for this case, and I do not see how one could complete the proof of the particular case of Fermat's theorem just mentioned.

Addition to the preceding memoir.

Since the preceding memoir was presented to the Academy, M. Legendre has published a second supplement to his *Theory of Numbers*, in which he proves the impossibility of the equation

$$x^5 + y^5 = z^5.$$

The case of the unknown divisible at the same time by 2 and by 5 is treated in that work as in the preceding memoir, and the author then proves the impossibility of the other case by means of a new analysis. The object of this addition is to establish two new theorems on indeterminate equations of the fifth degree, which include as a particular case Fermat's theorem for fifth powers. I arrive at them by starting from the results obtained above and by using an analysis that differs in several respects from that of M. Legendre and is entirely analogous to the method set out in the preceding memoir. We have seen that the success of the analysis used there is based on the fact that, in order to make the binomial $P^2 - 5Q^2$, in which Q must be divisible by 5, equal to an odd fifth power, it suffices to put

$$P + Q\sqrt{5} = (\varphi + \psi\sqrt{5})^5,$$

because this circumstance gives rise to the continual reproduction of the expression that we called the trinomial factor. In treating the new equations that form the object of this addition, one is likewise led to the binomial $P^2 - 5Q^2$, with Q still divisible by 5; but there is this difference: the numbers P and Q , which previously were one even and the other odd, are here both odd, and the

binomial $P^2 - 5Q^2$, which in the other case had to be an odd fifth power, must here be made equal to four times such a power. Now the formula that satisfies this latter condition in the most general manner can be made perfectly similar to the one that serves to satisfy the first condition, for I have observed that it can be presented as

$$P + Q\sqrt{5} = \frac{(\varphi + \psi\sqrt{5})^5}{2^4},$$

an expression distinguished from the result just recalled only in that the unknowns φ and ψ , instead of being one even and the other odd, must both be odd. Once this remark has been made, and once in treating the new equations now to occupy us one has arrived at the binomial that must be made equal to four times a fifth power, one sees at a glance that the impossibility of these equations should be proved by using a procedure altogether analogous to the course followed in the proof of Theorem V. We shall now enter into the matter, beginning by establishing the proposition already stated.

The numbers P and Q , the first of which is supposed not to be divisible by 5, being both odd and having no common divisor, the number $P^2 - 5Q^2$ will be of the form $8\lambda + 4$, and one may put

$$P^2 - 5Q^2 = 4L,$$

where L is odd and not divisible by 5. Multiplying this equation side by side with

$$3^2 - 5 \cdot 1^2 = 4,$$

one obtains

$$(3P \pm 5Q)^2 - 5(P \pm 3Q)^2 = 16L.$$

Since P and Q are both odd, it is evident that, by choosing the sign suitably, the expressions

$$\frac{3P \pm 5Q}{4}, \quad \frac{P \pm 3Q}{4}$$

will both be integers. Accordingly put

$$P' = \frac{3P \pm 5Q}{4}, \quad Q' = \pm \frac{P \pm 3Q}{4},$$

the sign outside the parentheses being chosen so as to give a positive value for Q' . The equation obtained above then becomes $P'^2 - 5Q'^2 = L$, and one easily verifies that

$$P + Q\sqrt{5} = (P' \pm Q'\sqrt{5})(3 \pm \sqrt{5}),$$

the signs being suitably chosen, and that the numbers P' and Q' are relatively prime and, moreover, one even and the other odd.

Suppose now that the number L is to be a fifth power. This condition is satisfied in the most general way by putting

$$P' + Q'\sqrt{5} = (M \pm N\sqrt{5})^5(9 + 4\sqrt{5})^p.$$

Substituting this value into the last equation gives

$$P + Q\sqrt{5} = (M \pm N\sqrt{5})^5(9 + 4\sqrt{5})^p(3 \pm \sqrt{5}),$$

in which the signs are independent, as in the two preceding equations. One can write $p = 5k + r$, with k a positive integer and r having one of the three values 0, 1, 2; the quantity $(9 \pm 4\sqrt{5})^p$ thus

decomposes into two factors, $(9 \pm 4\sqrt{5})^{5k}$ and $(9 \pm 4\sqrt{5})^{\pm r}$, of which the first can be omitted because it is absorbed into $(M \pm N\sqrt{5})^5$. If we observe moreover that, by virtue of

$$9 \pm 4\sqrt{5} = (9 \mp 4\sqrt{5})^{-1},$$

one can in $(9 \pm 4\sqrt{5})^{\pm r}$ change simultaneously the signs of r and of the radical, we may suppose r positive, and the equation above becomes

$$P + Q\sqrt{5} = (M \pm N\sqrt{5})^5 (9 \pm 4\sqrt{5})^r (3 \pm \sqrt{5}).$$

Since L must always be the fifth power of an odd number not divisible by 5, let us determine the necessary conditions for Q to be divisible by 5. Since the coefficient of $\sqrt{5}$ in the expansion of $(M \pm N\sqrt{5})^5$ is divisible by 5, while the rational part is not, M not being divisible by 5, one concludes, as in the proof of Theorem IV, that for Q to be divisible by 5, the coefficient of $\sqrt{5}$ in the developed value of $(9 \pm 4\sqrt{5})^r (3 \pm \sqrt{5})$ must be divisible by 5.

Substituting successively for r the three values 0, 1, 2, one finds that this occurs only when $r = 2$, the signs of the radicals in the two factors $(9 \pm 4\sqrt{5})^r$ and $(3 \pm \sqrt{5})$ being at the same time opposite. If one notes that

$$\frac{(3 \pm \sqrt{5})^3}{2^3} = 9 \pm 4\sqrt{5}, \quad 3 \mp \sqrt{5} = \frac{4}{3 \pm \sqrt{5}},$$

one finds that the preceding product is, in the case in question, equivalent to

$$\frac{(3 \pm \sqrt{5})^5}{2^4},$$

and substitution of this value in the equation obtained above changes it into

$$P + Q\sqrt{5} = \frac{(M \pm N\sqrt{5})^5 (3 \pm \sqrt{5})^5}{2^4}.$$

Since M and N are one even and the other odd, the numbers φ and ψ determined by

$$\varphi + \psi\sqrt{5} = (M \pm N\sqrt{5})(3 \pm \sqrt{5})$$

are both odd, and one has

$$P + Q\sqrt{5} = \frac{(\varphi + \psi\sqrt{5})^5}{2^4},$$

and consequently

$$P = \frac{\varphi(\varphi^4 + 2 \cdot 5^2 \varphi^2 \psi^2 + 5^3 \psi^4)}{2^4}, \quad Q = \frac{5\psi(\varphi^4 + 10\varphi^2 \psi^2 + 5\psi^4)}{2^4}.$$

For P and Q to be relatively prime, φ and ψ must have no common divisor and the first of these numbers must not be divisible by 5; conversely, if φ and ψ , the first supposed not divisible by 5, have no common divisor and are both odd, then the numbers P and Q will be integers and relatively prime. Indeed, the quarter of the quantity $\varphi^4 + 10\varphi^2 \psi^2 + 5\psi^4$ can be written as

$$\left(\frac{\varphi^2 + 5\psi^2}{2} \right)^2 - 5\varphi^2 \psi^2,$$

and since this last expression is evidently four times an odd number, the value of Q is integral and odd; the same is proved for P , and one easily verifies that P and Q , both odd, have no common divisor. We have therefore this theorem.

Theorem VII.

The numbers P and Q being relatively prime, both odd, and the latter divisible by 5, I say that to make the binomial $P^2 - 5Q^2$ equal to four times a fifth power with the proper generality, it suffices to put

$$P + Q\sqrt{5} = \frac{(\varphi + \psi\sqrt{5})^5}{2^4},$$

where the indeterminate numbers φ and ψ are relatively prime, both odd, and the first moreover not divisible by 5^7 .

Here is now the first of the new theorems announced at the beginning of this addition.

Theorem VIII.

Let the letter n denote a positive number other than 0 and 2, and let the number A be divisible neither by 2 nor by 5 nor by any prime number of the form $10k + 1$. It will be impossible to find two relatively prime numbers x and y such that

$$x^5 \pm y^5 = 5^n Az^5. \tag{\gamma}$$

The numbers x and y can be either both odd, or one even and the other odd. In the first case, z will be divisible by 2, and one may put $2z$ in place of z , which changes equation (γ) into

$$x^5 \pm y^5 = 2^5 5^n Az^5,$$

which is impossible since it evidently falls under Theorem V. It remains, then, to prove the impossibility of the second case, in which x and y are supposed one even and the other odd. If we put

$$x \pm y = p, \quad x \mp y = q,$$

then

$$2x = p + q, \quad \pm 2y = p - q,$$

and p and q are relatively prime and moreover both odd. Substituting the preceding values of $2x$ and $\pm 2y$ into equation (γ) , after multiplying its two members by 2^5 , gives

$$p(p^4 + 10p^2q^2 + 5q^4) = 2^5 5^n Az^5.$$

Since p must evidently be divisible by 5, put $p = 5r$, which gives

$$5^2 r(q^4 + 2 \cdot 5^2 q^2 r^2 + 5^3 r^4) = 2^5 5^n Az^5.$$

By hypothesis n is equal to unity or greater than 2. If n is equal to unity, then in the preceding equation z must be supposed divisible by 5. Therefore in that case one may put $5z$ in place of z , or, what is the same thing, give 5 the exponent 6, whence one sees that in all cases one may suppose $n > 2$.

⁷This theorem, like Theorem IV, has its analogues for many other prime numbers.

Putting the preceding equation in the form

$$r(q^4 + 2 \cdot 5^2 q^2 r^2 + 5^3 r^4) = 2^4 5^{n-2} A z^5,$$

and observing that $n > 2$ and q is prime to $p = 5r$, one sees that r must be supposed divisible by 5.

Choose now a positive number ν such that $n + \nu - 2$ is divisible by 5, and a number B that has no prime divisors except those of A and such that the product AB is a fifth power. Multiplying the last equation by $5^\nu B$, one gets

$$5^\nu B r (q^4 + 2 \cdot 5^2 q^2 r^2 + 5^3 r^4) = 2^4 5^{n+\nu-2} A B z^5.$$

Since all divisors of the trinomial factor, in which q and r are relatively prime, are of the form $10k + 1$, it is evident that it has no common divisor with B ; it is no less evident that it is also prime to $5^\nu r$, and consequently to $5^\nu B r$.

Since the factor $5^\nu B r$ and the trinomial factor are relatively prime, and since the first of these factors is odd, it follows from the last equation, whose second member is the product of 2^4 and the fifth power of an odd number, that $5^\nu B r$ is a fifth power, and the trinomial factor is a fifth power multiplied by 2^4 .

The quarter of the trinomial factor must therefore be four times a fifth power; and since this quarter can be written

$$\left(\frac{q^2 + 5^2 r^2}{2} \right)^2 - 5(5r^2)^2,$$

where the numbers $\frac{q^2 + 5^2 r^2}{2}$ and $5r^2$ are evidently relatively prime, both odd, and the latter moreover divisible by 5, it will suffice, by the theorem established at the beginning of this addition, to make the quarter of the trinomial factor equal to four times a fifth power by putting

$$\begin{aligned} \frac{q^2 + 5^2 r^2}{2} &= t \frac{t^4 + 2 \cdot 5^2 t^2 s^2 + 5^3 s^4}{2^4}, \\ 5r^2 &= 5s \frac{t^4 + 10t^2 s^2 + 5s^4}{2^4}. \end{aligned}$$

The indeterminate numbers t and s must be supposed relatively prime, both odd, and the first moreover not divisible by 5. Since t is not divisible by 5 and r is, as we saw above, it follows from the last of the preceding equations that s must be divisible by 5.

We saw above that $5^\nu B r$ had to be a fifth power. The number $5^{2\nu} B^2 r^2$, its square, must therefore be a power of the same degree, and even of the tenth degree. Multiplying both sides of the last equation by $2^4 5^{2\nu-1} B^2$, one obtains

$$2^4 5^{2\nu} B^2 r^2 = 5^{2\nu} B^2 s (t^4 + 10t^2 s^2 + 5s^4).$$

If then, for abbreviation, we put $2\nu = h$, $B^2 = C$, everything reduces to proving that it is impossible to find two relatively prime numbers t and s , both odd, with the latter s moreover divisible by 5, such that the product

$$(\delta) \quad 5^h C s (t^4 + 10t^2 s^2 + 5s^4)$$

is a fifth power multiplied by 2^4 .

It is easy to see that expression (δ) cannot be the product of 2^4 and a fifth power unless $5^h C s$ is a fifth power and the trinomial factor is a fifth power multiplied by 2^4 .

The quarter of the trinomial factor, having to be four times a fifth power, and being expressible in the form

$$\left(\frac{t^2 + 5s^2}{2}\right)^2 - 5(s^2)^2,$$

where the numbers $\frac{t^2 + 5s^2}{2}$ and s^2 are evidently relatively prime, both odd, and the latter moreover divisible by 5, is made equal to four times a fifth power by setting

$$\frac{t^2 + 5s^2}{2} = t' \frac{t'^4 + 2 \cdot 5^2 t'^2 s'^2 + 5^3 s'^4}{2^4},$$

$$s^2 = 5s' \frac{t'^4 + 10t'^2 s'^2 + 5s'^4}{2^4}.$$

The numbers t' and s' are supposed relatively prime, both odd, and the first t' moreover not divisible by 5. Since s is divisible by 5 and t' is not, it follows from the last equation that s' is also divisible by 5. One also concludes from the last equation that $\frac{2^5 \cdot 5}{16} s'^5 < s^2$, and hence that s' is much smaller than s .

The number $5^h C s$ having to be a fifth power, $5^{2h} C^2 s^2$, the square of the preceding number, must be a power of the same degree and even of the tenth degree. Multiplying both sides of the last equation by $2^4 5^{2h} C^2$, one obtains

$$2^4 5^{2h} C^2 s^2 = 5^{2h+1} C'^2 s' (t'^4 + 10t'^2 s'^2 + 5s'^4),$$

or, what is the same thing, putting $2h + 1 = h'$, $C^2 = C'$ in the second member,

$$(\delta') \quad 2^4 5^{2h} C^2 s^2 = 5^{h'} C' s' (t'^4 + 10t'^2 s'^2 + 5s'^4),$$

an equation whose two members must be fifth powers multiplied by 2^4 .

The product (δ') is perfectly similar to the product (δ) , and the number s' is much smaller than s ; hence, as in the proof of Theorem V of the preceding memoir, one concludes that the product (δ) cannot be equal to a fifth power multiplied by 2^4 , and consequently that equation (γ) cannot hold.

Fermat's theorem, in the case of fifth powers, is included as a particular case in the theorem just established. Indeed, since the equation $x^5 + y^5 = z^5$ cannot hold unless one of the unknowns, z for instance, is divisible by 5, we may put $5z$ in place of z ; this gives $x^5 + y^5 = 5^5 z^5$, an impossible equation because it falls under the last theorem.

A reasoning wholly similar to that by means of which we established Theorem VI from Theorem V can serve to deduce from the theorem just proved a new theorem, which can be stated as follows.

Theorem IX.

Let the number A be subject to the same restrictions as in the statement of Theorem VIII, and let this number give one of the eight remainders 3, 4, 9, 12, 13, 16, 21, 22 when divided by 25. It will be impossible to find two relatively prime numbers x and y such that $x^5 \pm y^5 = Az^5$.

ON LINEAR FORMS
IN WHICH ARE CONTAINED THE PRIME DIVISORS OF
CERTAIN
FORMULAE OF HIGHER DEGREES

DISSERTATION,

Composed by
GUSTAV LEJEUNE DIRICHLET,
Doctor of Philosophy,
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It is known from the doctrine of quadratic residues, or the theory of the divisors of formulae of the second degree, that the prime divisors of such formulae are contained in certain linear forms, and that there are linear forms distinct from these in which the non-divisors are comprised; hence, in order to settle the question whether a given prime divides a given formula or not, it is enough to examine whether the prime is contained in one or the other of the two preceding classes of forms. The matter is quite otherwise for formulae of higher degree: their divisors cannot be distinguished from their non-divisors in this way, and criteria must be used that are quite different from those proposed in the doctrine of quadratic residues.

To illustrate by an example the manner in which the divisors of formulae of higher degree are distinguished from the non-divisors, let us consider the formula of the fourth degree $x^4 - 3$. From the simplest principles of higher arithmetic it follows that, for this formula, only primes of the form $12n + 1$ require a criterion of a higher kind; for the remaining primes the question can be decided by means of the familiar theory of quadratic residues. If a prime of this form is proposed and it is asked whether it is a divisor of $x^4 - 3$ or not, the following criterion is to be used. Reduce the prime to the form $t^2 + 3u^2$, which, as is known, can always be done in exactly one way. Then the number t , which is evidently divisible neither by 2 nor by 3, will be of the form $12n \pm 1$, or of the form $12n \pm 5$. If the former case occurs, the prime will be a divisor of $x^4 - 3$; if the latter, it will not be.

There is another criterion suitable for deciding the same question; but in order that it have a more elegant form, it is to be referred not to $x^4 - 3$, but to $x^4 + 3$. The latter is so closely connected with the former that, if we know whether a certain number is or is not a divisor of the latter, we can at once conclude whether the same number divides the former. To use this second criterion, the proposed prime number is to be resolved into two squares. Of the squares supplied by this decomposition, one will plainly be even and the other odd; and it is no less easy to see that one of these two squares, and only one, will be divisible by 3. With these preliminaries, the criterion may be stated as follows:

If the even square is divisible by 3, the prime will divide the formula $x^4 + 3$; if the odd one is divisible by 3, it will not divide it.

The proofs of the theorems just stated will be found in several memoirs in which I shall attempt to construct the general theory of the divisors of the formula $\alpha x^4 + \beta x^2 + \gamma$, the first of which has lately appeared in the celebrated mathematical journal of Crelle under the title *Recherches sur les diviseurs premiers d'une classe de formules du quatrième degré. Premier Mémoire*. The preceding theorems have been brought forward here only so that an example may show how much formulae of higher degree differ from quadratic formulae when the nature of their prime divisors is investigated.

Although this difference appears to hold in general, there are nevertheless particular formulae of almost any degree which, with respect to their divisors, behave in a way similar to formulae of the second degree, so that from the linear form in which a prime is contained one can know whether it divides a formula of this kind. To this class belong the formulae comprised in the expression $x^n \pm 1$, on which Euler carried out investigations no less remarkable for the exceptional simplicity of his reasoning than for the great generality of the theorems established by it. In pursuing these investigations of Euler I came upon a certain new species of formulae which, as far as divisors are concerned, has similar properties, as I shall explain below.

Since part of the following investigation is based on some of the Eulerian theorems just mentioned, I have thought it useful to place before the reader the theorems that will be needed. I omit the proofs for brevity, referring the reader to Euler's dissertations or to Legendre's excellent work, which expounds both these and the other arithmetical discoveries of Euler with the greatest clarity.

Theorem I. If p is an odd prime, all the prime divisors of the formula

$$\frac{x^p - 1}{x - 1}$$

are contained in the linear form $2mp + 1$, and conversely every prime contained in this form is a divisor of the formula.

Theorem II. If α is a power of the number 2, the divisors of the formula $x^\alpha + 1$ are contained in the form $2m\alpha + 1$, and conversely every prime of this form divides the formula.

The formulae whose divisors we shall investigate below arise from the expansion of the power $(x + \sqrt{b})^n$, where n and b are given integers, the latter not a square, while x denotes an indeterminate integer. Suppose that, when the expansion is completed, it gives $U + V\sqrt{b}$, where both U and V are assumed free of the irrational quantity $\sqrt{b}$; thus we have

$$(x + \sqrt{b})^n = U + V\sqrt{b}, \quad (x - \sqrt{b})^n = U - V\sqrt{b},$$

$$V = nx^{n-1} + \frac{n(n-1)(n-2)}{1 \cdot 2 \cdot 3}x^{n-3}b + \frac{n(n-1)(n-2)(n-3)(n-4)}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5}x^{n-5}b^2 + \dots$$

Our object is to determine the prime numbers by which V becomes divisible when all integral values, positive as well as negative, are successively assigned to x . Only values not prime to b are excepted, since they disturb the neatness of the theorems. It is also very easy to see that the case in which x is assumed to have a common divisor with b can be reduced to the case in which x and b are relatively prime.

To begin the investigation, let us show that U and V can have no common divisor, except 2, whenever x takes a value prime to b . To prove this, we start from the contrary supposition. Let δ be a prime dividing both U and V . If the first of the equations is multiplied by the second, we get

$$(x^2 - b)^n = U^2 - bV^2,$$

from which it follows that δ divides $x^2 - b$. Using the notation introduced by Gauss, we write this as

$$x^2 \equiv b \pmod{\delta},$$

and from this congruence the following are obtained:

$$x^4 \equiv b^2, \quad x^6 \equiv b^3, \quad x^8 \equiv b^4, \quad \dots \pmod{\delta}.$$

If now in the expression for V we substitute, in place of $b, b^2, b^3, \dots$, the values congruent to them modulo δ , namely $x^2, x^4, x^6, \dots$, we obtain

$$V \equiv x^{n-1} \left(n + \frac{n(n-1)(n-2)}{1 \cdot 2 \cdot 3} + \frac{n(n-1)(n-2)(n-3)(n-4)}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5} + \dots \right) \pmod{\delta}.$$

Since the expression in parentheses is plainly

$$\frac{1}{2}((1+1)^n - (1-1)^n),$$

that is, 2^{n-1} , we may also write

$$V \equiv 2^{n-1} x^{n-1} \pmod{\delta}.$$

It follows from this congruence that the odd prime δ , which we supposed to divide V , also divides x . But we saw above that $x^2 - b$ is divisible by δ ; hence δ would divide both x and b , contrary to the supposition that x and b are relatively prime. Therefore the supposition that U and V have an odd common divisor cannot hold.

With these preliminaries, let us inquire into the nature of the prime divisors of the formula V . Although the method to be used here can be applied for any value of n , in order that this dissertation not become too long I shall examine only two cases: first, when n is an odd prime, and second, when n is a power of 2.

Let n first be an odd prime, denoted by p , and let k be an odd prime different from p which divides the formula V . Two cases are now to be distinguished, according as b is a quadratic residue or a quadratic non-residue modulo k .

In the former case there is a number μ satisfying $\mu^2 \equiv b \pmod{k}$. The prime k , which we have assumed to divide V , will divide the same formula if we substitute for b the value μ^2 , congruent to it modulo k . Comparing the first and second equations, the formula V can be written

$$\frac{(x + \sqrt{b})^p - (x - \sqrt{b})^p}{2\sqrt{b}}.$$

After the substitution just described this becomes

$$\frac{(x + \mu)^p - (x - \mu)^p}{2\mu}.$$

I say that in this expression neither $x + \mu$ nor $x - \mu$ is divisible by k . For if one were, and the other were not, then $(x + \mu)^p - (x - \mu)^p$ would not be divisible by k . That both cannot be divisible by k is shown as follows: then k would divide each of

$$\frac{1}{2}((x + \mu) + (x - \mu)) \quad \text{and} \quad \frac{1}{2}((x + \mu) - (x - \mu)),$$

that is, both x and μ ; and since $\mu^2 \equiv b \pmod{k}$, k would also divide b . The numbers x and b would therefore have the common divisor k , against our assumption.

Since neither $x + \mu$ nor $x - \mu$ is divisible by k , the congruence

$$(x - \mu)y \equiv x + \mu \pmod{k}$$

can be satisfied by some number y . If now, in the value of $2\mu V$, we substitute for $x + \mu$ the congruent number $y(x - \mu)$, we have the following expression divisible by k :

$$(x - \mu)^p(y^p - 1).$$

Since $x - \mu$, as was seen above, is not divisible by k , it follows that $y^p - 1$ is a multiple of k . I next say that $y - 1$ is not divisible by k : for if k divided $y - 1$, we would have $y \equiv 1 \pmod{k}$, and the congruence $(x - \mu)y \equiv x + \mu \pmod{k}$ would become $2\mu \equiv 0 \pmod{k}$; that is, μ , and hence b , would be divisible by k . But inspection of the formula V shows that k cannot divide b , unless either k is equal to p or k divides x , both of which cases were excluded above. It follows, by Theorem I, that the prime k is of the form $2mp + 1$.

We have therefore the theorem: "No prime for which b is a residue can divide the formula V , unless it is contained in the form $2mp + 1$." This proposition is also conversely true, and it can be proved that every prime of the form $2mp + 1$ is a divisor of V . Since the proof of this proposition offers no difficulty, I omit it for brevity.

We now pass to the primes for which b is a non-residue, and whose relation to the formula V , in so far as they are divisors or non-divisors of the formula, is determined by the following theorems:

If a prime for which b is a non-residue is at the same time of the form $2mp - 1$, then it is a divisor of V .

No prime k for which b is a non-residue can divide V , unless it is contained in the form $2mp - 1$.

The former of these theorems was proposed and proved long ago by the illustrious Lagrange in a memoir preserved in the academic collection, *Nouveaux Mémoires de Berlin*, année 1775; with its help he established several particular propositions belonging to the doctrine of quadratic residues. The theorem and the demonstration by which that great man supported it are also found in the excellent work *Disquisitiones arithmeticae*; and since that work is in the hands of everyone who works in Diophantine analysis, I have not thought it necessary to add the proof here.

The latter theorem, which is the converse of the former, has, so far as I know, not yet been stated by anyone. Its less obvious proof is carried through by reasonings very similar to those which I have already used elsewhere, in the *Mémoire sur l'impossibilité de quelques équations du cinquième degré*, in an entirely different argument.

Consider the formula $(x + \sqrt{b})^{k+1}$, and suppose that its expansion gives $M + N\sqrt{b}$, where M and N are free of the irrational quantity $\sqrt{b}$. For brevity I do not write out the value of N obtained by substituting $k + 1$ for n in the expression for V . It is clear that the coefficients of all terms of N are divisible by k , except the coefficients of the first and last terms. Since these terms are $(k + 1)x^k$ and $(k + 1)xb^{(k-1)/2}$, we get

$$N \equiv (k + 1)(x^k + xb^{\frac{k-1}{2}}) \pmod{k}.$$

The number b , which we have supposed to be a non-residue modulo k , satisfies

$$b^{\frac{k-1}{2}} \equiv -1 \pmod{k},$$

and since we also have $x^k \equiv x \pmod{k}$, it follows that

$$x^k + xb^{\frac{k-1}{2}} \equiv 0 \pmod{k}.$$

Thus N will be divisible by k whatever value is assigned to x .

Now suppose that, for a determined value of x , the formula V becomes divisible by a prime k for which b is a non-residue; we shall try to prove that such a prime is of the form $2mp - 1$. Since V is divisible by k , the formula U will not be divisible by k , because we have seen that it is prime to V , provided x is assigned a value prime to b , as we always assume. It remains to prove that the prime k is of the form $2mp - 1$, or that $k + 1$ is divisible by p .

If $k+1$ were not divisible by p , there would exist positive integers g and h such that $g(k+1) - hp = 1$. Return now to the formula $M + N\sqrt{b}$, in which we saw N to be divisible by k , and raise it to the g -th power. If this power is denoted by $M' + N'\sqrt{b}$, it is easy to see that N' is divisible by N , and therefore by k ; but M' is not divisible by k , since $M' + N'\sqrt{b}$ is a power of $x + \sqrt{b}$, with exponent $g(k+1)$, and consequently M' and N' cannot have a common divisor. In the same way, if $(U + V\sqrt{b})^h = U' + V'\sqrt{b}$, then V' will be divisible by k , but U' will not. Comparing the equations

$$(x + \sqrt{b})^{g(k+1)} = M' + N'\sqrt{b}, \quad (x + \sqrt{b})^{hp} = U' + V'\sqrt{b}, \quad g(k+1) - hp = 1$$

gives

$$(x + \sqrt{b})(U' + V'\sqrt{b}) = M' + N'\sqrt{b}.$$

Multiplying out and equating separately the rational parts and the coefficients of $\sqrt{b}$, we obtain

$$M' = U'x + bV', \quad N' = V'x + U'.$$

The latter of these equations is plainly impossible, since N' and V' are multiples of k , while U' is not divisible by k . Hence the supposition from which we started cannot hold, and $k+1$ must indeed be divisible by p ; equivalently, k is of the form $2mp - 1$, as was to be shown.

If what has been proved so far is joined with the familiar doctrine of quadratic residues, one can assign the linear forms in which the prime divisors of the formula V are included.

For illustration, let $p = 5$ and $b = -1$. In this case V is

$$5x^4 - 10x^2 + 1.$$

From what we have proved above, primes for which -1 is a residue cannot be divisors of V unless they are of the form $10m + 1$; and by the theory of quadratic residues, primes for which -1 is a residue are contained in the form $4n + 1$. The numbers included in both forms $10m + 1$ and $4n + 1$ are of the form $20h + 1$. Thus every prime for which -1 is a residue and which at the same time divides V is contained in that form. As for the prime divisors of V for which -1 is a non-residue, one similarly finds that all such primes are comprised in the form $20h - 1$. Therefore the double formula $20h \pm 1$ supplies all primes dividing the expression $5x^4 - 10x^2 + 1$, and conversely every prime contained in either of the forms $20h + 1$, $20h - 1$ is a divisor of V .

As a second example take $p = 7$, $b = 2$. Then

$$V = 7x^6 + 70x^4 + 84x^2 + 8.$$

Primes for which 2 is a residue are contained in the forms $8m + 1$, $8m + 7$; but such primes cannot divide V unless they are of the form $14n + 1$. Comparing these forms with the preceding ones

gives the forms $56h + 1, 56h + 15$, in which are included the prime divisors of V for which 2 is a residue. Prime divisors of V for which 2 is a non-residue are found to be comprised in the forms $56h + 13, 56h + 27$. The formulae $56h + 1, 13, 15, 27$ therefore supply all primes dividing the quadrimomial $7x^6 + 70x^4 + 84x^2 + 8$, and conversely every prime contained in one of these forms is a divisor of the quadrimomial.

There are special cases which deserve particular attention because of the elegance of the theorems. I shall now treat more fully those cases that occur whenever b is equal to p , taken positively or negatively.

First let $b = p$, and suppose that the prime p is of the form $4n + 1$. It is known that the primes for which $b = p$ is a residue are contained in $\frac{1}{2}(p - 1)$ linear forms of the following kind:

$$mp + 1, \quad mp + a, \quad mp + a', \quad mp + a'', \dots,$$

where the numbers $1, a, a', a'', \dots$, which may be assumed positive and less than p , are all distinct and arise from the remainders of the squares

$$1, 2^2, 3^2, \dots, \left(\frac{1}{2}(p - 1)\right)^2$$

on division by the prime p . Among these numbers, $p - 1$ is known to occur whenever p is of the form $4n + 1$. The primes for which b is a non-residue are contained in $\frac{1}{2}(p - 1)$ other linear forms:

$$mp + \beta, \quad mp + \beta', \quad mp + \beta'', \dots,$$

where $\beta, \beta', \beta'', \dots$, assumed less than p , are distinct from each other and from $1, a, a', a'', \dots$, so that $p - 1$ cannot occur among $\beta, \beta', \beta'', \dots$.

We proved above that a prime for which b is a residue, and hence is contained in one of the first forms, cannot be a divisor of V unless it is of the form $mp + 1$, which is the same as the first of those forms and cannot agree with any of the others. It follows that primes for which $b = p$ is a residue and which also divide V are comprised in the form $mp + 1$, and conversely, and so on. As for the primes for which $b = p$ is a non-residue, and which are therefore contained in one of the second forms, what was established above shows that such primes cannot be divisors of V unless at the same time they are included in the form $mp - 1$, or equivalently in $mp + p - 1$. But this form cannot agree with any of the second forms, since none of $\beta, \beta', \beta'', \dots$ is equal to $p - 1$. It is therefore proved that no prime for which $b = p$ is a non-residue is a divisor of V , and that all primes dividing V are comprised in the form $mp + 1$.

If the case $b = -p$ is examined in the same way, with p denoting a prime of the form $4m + 1$, one finds that no prime can be a divisor of V unless it is contained in one of the forms $4mp \pm 1$; and conversely every prime included in one of these forms really divides V .

The third case occurs when $b = p$ and p is of the form $4m + 3$. In this case the primes dividing V are found to be comprised in the forms $4mp + 1$ and $4mp + 2p - 1$.

It remains to examine the fourth case. Here p is of the form $4m + 3$, and b is equal to p taken negatively. If the considerations used in the preceding cases are applied to the present case, we arrive at this theorem:

No prime can divide the formula V , in which $b = -p$ and p denotes a prime of the form $4m + 3$, unless it is contained in one of the forms $mp + 1, mp - 1$; and conversely all primes comprised in these forms are divisors of V .

Whenever the fourth case occurs, the formula V is found to be the product of two rational factors. We must now inquire more deeply into their nature, in order to decide of which factor a prime dividing the formula is a divisor.

It is known from the theory of the division of the circle, due to the illustrious Gauss, that the expression

$$4 \frac{t^p - u^p}{t - u},$$

whenever p is a prime of the form $4n + 3$, can be reduced to the form

$$R^2 + pS^2,$$

where R and S are integral rational functions. The formula $4 \frac{t^p - u^p}{t - u}$ is plainly a symmetric function of t and u ; and from the proof by which that great man established the theorem just mentioned, it follows easily that R is a function which takes the opposite value when t and u are interchanged.

Since S is a symmetric function of degree $\frac{p-1}{2}$, it will be a sum of several terms of the form

$$at^m u^{k-m} + at^{k-m} u^m,$$

where $k = \frac{p-1}{2}$ and one may suppose $m < \frac{k}{2}$. Such a term can also be written as

$$at^m u^m (t^{k-2m} + u^{k-2m}).$$

If in this expression we substitute for t and u respectively $x + \sqrt{-p}$ and $x - \sqrt{-p}$, then $t^m u^m$ takes the value $(x^2 + p)^m$, which is an even function of x (by this term, for brevity, I mean a function in which all exponents are even). The other factor, $t^{k-2m} + u^{k-2m}$, since the exponent $k - 2m$ is odd, will by the same substitution be transformed into an integral odd function of x , that is, into a function in which all exponents are odd. Thus the value of that term after substitution will be an odd function of x . Since what has just been said holds for every term from which S is built, it follows that S , after the same substitution, will be changed into an odd function of x . This function, all of whose coefficients are easily seen to be even, will be denoted by $2S'$.

In a similar way one finds that the formula R , if $x + \sqrt{-p}$ and $x - \sqrt{-p}$ are substituted respectively for t and u , will take a value consisting of two factors, of which the first is $\sqrt{-p}$ and the second is a rational even function of x . If the latter is denoted by $2R'$, then $2R'\sqrt{-p}$ is the value that R obtains under the substitution.

If now in the formula $R^2 + pS^2$ we substitute $2R'\sqrt{-p}$ and $2S'$ for R and S , respectively, the same value is obtained as that which $4 \frac{t^p - u^p}{t - u}$ takes when $x + \sqrt{-p}$ and $x - \sqrt{-p}$ are substituted for t and u . Thus we have

$$2 \frac{(x + \sqrt{-p})^p - (x - \sqrt{-p})^p}{\sqrt{-p}} = 4pS'^2 - 4pR'^2.$$

Since also

$$V = \frac{(x + \sqrt{-p})^p - (x - \sqrt{-p})^p}{2\sqrt{-p}},$$

we conclude that

$$V = p(S' + R')(S' - R').$$

It is therefore proved that V/p can be decomposed into two rational factors whenever $b = -p$ and the prime p is of the form $4n + 3$.

Since R' is an even function and S' an odd function of x , it is clear that when the sign of x is changed the former factor $S' + R'$ changes into $R' - S'$, that is, into the negative of the latter factor; and conversely the latter factor $S' - R'$, after the same change, becomes $-S' - R'$, that is, the negative of the former factor.

Having established this, let k be a prime divisor of the formula V , and let α be a value to be assigned to x so that V becomes divisible by k . I now say that $R' + S'$ will be divisible by k if x is given a suitable value. For since k divides the formula

$$V = p(R' + S')(R' - S'),$$

with $x = \alpha$, either $R' + S'$ or $R' - S'$ will be divisible by k . If the former case occurs, α is the required value; if the latter occurs, it follows from the preceding observations that $R' + S'$ will be divisible by k when x is made equal to $-\alpha$. Thus in this case the suitable value is $-\alpha$. In the same way it is shown that a value of x can be chosen so that $R' - S'$ is divisible by k . Hence every prime dividing V is a divisor of each of the factors $R' + S'$, $R' - S'$, where it hardly needs to be noted that these factors are not divisible by k for the same value of x , but for opposite values, that is, for values differing only in sign.

To illustrate the preceding discussion by a few examples, first put $p = 7$. In this case the factors of V are found to be

$$x^3 - 7x^2 + 7x + 7, \quad x^3 + 7x^2 + 7x - 7,$$

which are plainly not different from one another with respect to divisors. All prime divisors of these formulae are contained in the form $7n \pm 1$, and conversely every prime comprised in this form is a divisor of the formulae.

Secondly, let $p = 11$. The factors of the fifth degree whose product is V in this case are

$$\begin{aligned} x^5 + 11x^4 - 2 \cdot 11x^3 - 2 \cdot 11^2x^2 - 3 \cdot 11^2x - 11^2, \\ x^5 - 11x^4 - 2 \cdot 11x^3 + 2 \cdot 11^2x^2 - 3 \cdot 11^2x + 11^2. \end{aligned}$$

Neither of these formulae is divisible by a prime unless that prime is contained in the form $11n \pm 1$, and conversely every prime, etc.

Up to this point we have supposed that the number n in the formulae is an odd prime. We now consider the case in which n is a power of the number 2. The theorems belonging to this case are to be referred not to the formula V , but to the formula U . In what follows we shall denote the number n by α , and k will be an odd prime dividing the formula U . Again two cases are to be distinguished, according as b is a quadratic residue or non-residue modulo k .

If b is a residue modulo k , there is a number μ such that $\mu^2 \equiv b \pmod{k}$. The prime k , which we have assumed to be a divisor of U , will divide the same formula if μ^2 is substituted for b . The formula $2U$, which, as the comparison of the first and second equations shows, can also be written

$$2U = (x + \sqrt{b})^\alpha + (x - \sqrt{b})^\alpha,$$

is changed by the indicated substitution into $(x + \mu)^\alpha + (x - \mu)^\alpha$. This expression plainly cannot be divisible by k unless k divides either both or neither of $x + \mu$ and $x - \mu$. That the former case cannot occur is shown as follows. Then each of the numbers

$$\frac{1}{2}(x + \mu) + \frac{1}{2}(x - \mu), \quad \frac{1}{2}(x + \mu) - \frac{1}{2}(x - \mu)$$

would be divisible by k , that is, both x and μ would be divisible by k ; and since $b \equiv \mu^2 \pmod{k}$, k would also divide b , contrary to the supposition that the values assigned to x are prime to b .

Since neither $x + \mu$ nor $x - \mu$ is divisible by k , a number y can be assigned satisfying the congruence

$$(x + \mu)y \equiv x - \mu \pmod{k}.$$

If now in the formula $2U$ we substitute for $x - \mu$ the value congruent to it modulo k , namely $y(x + \mu)$, we obtain this expression divisible by k :

$$(x + \mu)^\alpha (y^\alpha + 1).$$

Therefore the prime k must divide the factor $y^\alpha + 1$, and by Theorem II it follows that k is of the form $2m\alpha + 1$. Thus we have the theorem: "All primes for which b is a residue and which divide the formula U , where $n = \alpha$ is a power of the number 2, are contained in the form $2m\alpha + 1$." This proposition is also conversely true and can be proved in a similar way.

We pass to the consideration of primes for which b is a quadratic non-residue and for which the following theorem holds: "If a prime for which b is a non-residue is of the form $2m\alpha - 1$, then it is a divisor of U ; and conversely no prime for which b is a non-residue divides the formula unless it is contained in the form $2m\alpha - 1$." Since the proof of the first part of this theorem is sufficiently similar to what we have set out above, I leave it to the reader to develop. The proof of the latter part, however, requires new devices and will be explained more fully.

Let k be a prime for which b is a non-residue and which divides U for a determined value of x . We must prove that k is of the form $2m\alpha - 1$, or that $k + 1$ is divisible by 2α . Suppose that $k + 1$ is not divisible by 2α , and let us see what follows. Let β denote the highest power of 2 dividing $k + 1$. Then

$$\frac{k + 1}{\beta} \text{ is an odd number and } \frac{2\alpha}{\beta}$$

is either 2 or a power of 2. Therefore the numbers $\frac{k+1}{\beta}$ and $\frac{2\alpha}{\beta}$ have no common divisor, whence positive integers can be assigned such that

$$g \frac{k + 1}{\beta} - \frac{2\alpha}{\beta} h = 1.$$

This equation may also be written

$$g(k + 1) - 2h\alpha = \beta.$$

We saw above that N in the equation $M + N\sqrt{b} = (x + \sqrt{b})^{k+1}$ is divisible by k for every value of x . If $M + N\sqrt{b}$ raised to the power g is denoted by $M' + N'\sqrt{b}$, then N' is divisible by N and therefore by k , while M' is not divisible by k , since $M' + N'\sqrt{b}$ is plainly a power of $x + \sqrt{b}$; hence M' and N' cannot have an odd common divisor.

We have the equation

$$(x + \sqrt{b})^\alpha = U + V\sqrt{b}.$$

Here, since U is divisible by k , the formula V will not be, for it is prime to U . From the preceding equation we get

$$(x + \sqrt{b})^{2\alpha} = U^2 + bV^2 + 2UV\sqrt{b}.$$

Equivalently, if for brevity we put $U^2 + bV^2 = P$ and $2UV = Q$, then

$$(x + \sqrt{b})^{2\alpha} = P + Q\sqrt{b}.$$

Since k does not divide b , and since U is a multiple of k while V is not, it follows that k divides Q but not P . Now if we put

$$(P + Q\sqrt{b})^h = P' + Q'\sqrt{b},$$

then, as above, it can be proved that Q' will be a multiple of k , while P' will not.

From comparison of the equations

$$(x + \sqrt{b})^{g(k+1)} = M' + N'\sqrt{b}, \quad (x + \sqrt{b})^{2h\alpha} = P' + Q'\sqrt{b}, \quad g(k+1) - 2h\alpha = \beta,$$

we get

$$M' + N'\sqrt{b} = (P' + Q'\sqrt{b})(x + \sqrt{b})^\beta.$$

Putting, for brevity, $(x + \sqrt{b})^\beta = R + S\sqrt{b}$, this may also be written

$$M' + N'\sqrt{b} = (P' + Q'\sqrt{b})(R + S\sqrt{b}).$$

Multiplying and equating separately the rational parts and the coefficients of $\sqrt{b}$, it follows that

$$M' = P'R + bQ'S, \quad N' = P'S + Q'R.$$

Since Q' and N' are multiples of k , but P' is not divisible by k , the latter of the preceding equations plainly cannot hold unless S is divisible by k .

We have the equations

$$(x + \sqrt{b})^\alpha = U + V\sqrt{b}, \quad (x + \sqrt{b})^\beta = R + S\sqrt{b},$$

and comparison gives the new relation

$$U + V\sqrt{b} = (R + S\sqrt{b})^{\alpha/\beta}.$$

Here α/β is an integer, since $2\alpha/\beta$ is either 2 or a power of 2. It follows from this equation, in which the exponent α/β is integral, that V is divisible by S . And since k , as we saw above, divides the formula S , one would have to conclude that k is a divisor of V , contrary to what was established above. For we showed that the formulae U and V cannot have an odd common divisor; hence the prime k , which we assumed to be a divisor of U , cannot divide V . Thus it is proved that the supposition from which we started cannot hold; $k+1$ is in fact divisible by 2α , or k is of the form $2m\alpha - 1$.

Here are some examples belonging to the theorems just proved. Let $\alpha = 4$. Then U is

$$x^4 + 6bx^2 + b^2.$$

This formula will be divisible by no prime for which b is a quadratic residue unless that prime is contained in the form $8m+1$, and by no prime for which b is a non-residue unless it is included in the form $8m-1$.

Put $b = -3$. Then U is

$$x^4 - 18x^2 + 9.$$

From the theory of quadratic residues it is known that primes for which -3 is a residue are of the form $3m+1$, while primes for which -3 is a non-residue are of the form $3m-1$. Combining this with the preceding result, one finds: "the formula $x^4 - 18x^2 + 9$ is divisible by no prime unless that prime is contained in one of the forms $24m \pm 1$, and conversely every prime, etc."

Another example is obtained by taking $\alpha = 8, b = 5$. In this case the value of the formula U is

$$x^8 + 140x^6 + 1750x^4 + 3500x^2 + 625.$$

The prime divisors of this expression will be of the form $16m + 1$ or of the form $16m - 1$, according as 5 is a quadratic residue or non-residue of them. Combining this correctly with the known proposition that primes for which 5 is a residue are of the form $5m \pm 1$, while primes for which 5 is a non-residue are of the form $5m \pm 2$, gives the theorem:

The primes dividing $x^8 + 140x^6 + 1750x^4 + 3500x^2 + 625$ are contained in the forms $80m + 1, 47, 49, 63$, and conversely every prime comprised in one of these forms is a divisor of the formula.

RESEARCHES ON THE PRIME DIVISORS OF A CLASS OF FORMULAE OF THE FOURTH DEGREE.

First memoir.

In the literary announcements of Göttingen (11 April 1825) one finds the extract of a memoir of indeterminate analysis which Mr. Gauss presented to the Royal Society of Sciences of that city, but which has not yet been printed. This memoir is the first of a series of memoirs which the illustrious author of the *Disquisitiones arithmeticae* proposes to give on the theory of biquadratic residues, and its object is to determine the distinctive characters of the prime divisors of the formula $x^4 - 2$. The author establishes in it two extremely elegant theorems which can serve to decide whether a prime number, divisor of $x^2 - 2$, divides or does not divide the preceding formula. Having learned, in the course of the year just ended, of the cited extract, which contains only the statements of the two theorems just mentioned and of some auxiliary propositions, I wished to demonstrate on my side the beautiful theorems discovered by Mr. Gauss. The researches which I made with this aim led me to find a proof founded on extremely simple considerations and probably entirely different from that of Mr. Gauss, which appears to require very delicate and rather extensive preliminary researches. I then applied analogous considerations to other questions and particularly to the search for the properties which distinguish the prime divisors of the formula $\alpha x^4 + \beta x^2 + \gamma$; and I thus arrived at a large number of interesting theorems. The rapid exposition of part of the results to which these researches have led me is the object of the present memoir. I begin by setting down some definitions and by stating some theorems very easy to establish and on which we shall have to rely in what follows.

1.

"If one can assign to the indeterminate x a value such that $x^4 - A$ becomes divisible by B , A will be called a biquadratic residue with respect to B ."

It is easy to see that, if a number A is a biquadratic residue with respect to a number B , or in other words, if B is a divisor of $x^4 - A$, each prime factor of B will likewise be a divisor of $x^4 - A$, and reciprocally, that if this condition holds with respect to every prime factor of the number B , B will itself be a divisor of $x^4 - A$. Thus, when it is a question of assigning all the numbers which divide the formula $x^4 - A$, one can confine oneself to considering prime numbers.

It is no less evident that, for a number to divide the formula $x^4 - A$, it is necessary that this number be a divisor of $x^2 - A$. There are therefore only the prime divisors of this latter formula to examine, that is to say, the prime numbers with respect to which A is a quadratic residue. One may add that, relative to those among these latter which are of the form $4n + 3$, the question presents no difficulty; for one verifies by a very simple reasoning that every prime number $4n + 3$, divisor of $x^2 - A$, also divides the formula $x^4 - A$.

Let p be a prime number $4n + 1$, divisor of $x^2 - A$, A designating a positive

or negative number not divisible by p ; one has, as is known,

$$A^{\frac{p-1}{2}} \equiv 1 \pmod{p}.$$

From this one concludes

$$A^{\frac{p-1}{4}} \equiv \pm 1 \pmod{p},$$

and one proves easily that the upper or lower sign will hold according as p divides or does not divide the formula $x^4 - A$. Thus one may state this theorem:

" A designating a number which is a quadratic residue with respect to the prime number $p = 4n + 1$, one will have

$$A^{\frac{p-1}{4}} \equiv 1 \quad \text{or} \quad A^{\frac{p-1}{4}} \equiv -1 \pmod{p}.$$

In the first case, A will be a biquadratic residue with respect to p ; in the second, A will be a biquadratic non-residue with respect to p ."

If this theorem is applied to the case where $A = -1$, one will find that -1 is a biquadratic residue with respect to the prime numbers $8n + 1$, and on the contrary a non-residue relative to those of the form $8n + 5$. From the preceding theorem one easily deduces the following other theorem, in whose statement one supposes, as in all that follows, that p is a prime number $4n + 1$, and that A and A' designate numbers not divisible by p and which are both quadratic residues with respect to p .

"If the numbers A and A' are both biquadratic residues or both biquadratic non-residues with respect to p , the product AA' will be a biquadratic residue with respect to p ; if, on the contrary, one of the numbers A and A' is a residue and the other a biquadratic non-residue relative to p , AA' will be a biquadratic non-residue with respect to p ."

Taking $A' = -1$, and having regard to what precedes, one will see that, if p is of the form $8n + 1$, A will at the same time as $-A$ be a biquadratic residue or non-residue, and that, on the contrary, if p is of the form $8n + 5$, one of the numbers A and $-A$ will be a residue and the other a biquadratic non-residue with respect to p .

2.

After establishing these preliminaries, we shall deal with the search for the characters which distinguish the prime divisors of the formula $x^4 - 2$. It is known that the prime divisors of $x^2 - 2$ are of one of these two forms: $8n + 1$, $8n + 7$, and that reciprocally every prime number of one of these forms divides the formula $x^2 - 2$. According to what we said in the preceding paragraph, we need not take account of the latter of these two forms, and it will suffice to consider the prime numbers $8n + 1$. Let p be a prime number of this kind, and put, as one is permitted to do,

$$p = t^2 + 2u^2,$$

where u will be even and t odd. Let $u = 2^{\nu} k k' k'' \dots$, 2^{ν} being the highest power of 2 which divides u , and $k, k', k'', \dots$ designating odd prime numbers, several of which may be equal to one another. The equation $t^2 + 2u^2 = p$ immediately gives

$t^2 \equiv p \pmod{k}$. The number p is therefore a quadratic residue with respect to k , which we shall write thus:

$$\left(\frac{p}{k}\right) = 1,$$

adopting the notation used by Mr. Legendre. Recall that, if c designates a prime number and M any number not divisible by c , this illustrious geometer uses the sign $\left(\frac{M}{c}\right)$ to designate the remainder which one obtains by dividing by c the power $M^{\frac{c-1}{2}}$, a remainder which is known to be equal to 1 or -1 , according as M is or is not a quadratic residue with respect to c . Applying to the relation $\left(\frac{p}{k}\right) = 1$ the theorem known under the name law of reciprocity (the *theorema fondamentale* of Mr. Gauss), one will have, p being of the form $4n + 1$,

$$\left(\frac{k}{p}\right) = 1.$$

One finds in the same way:

$$\left(\frac{k'}{p}\right) = 1, \quad \left(\frac{k''}{p}\right) = 1, \quad \text{etc.}$$

On the other hand, since p is of the form $8n + 1$, one also has $\left(\frac{2}{p}\right) = 1$, and consequently $\left(\frac{2^\nu}{p}\right) = 1$. Multiplying this last relation by all the preceding ones, one obtains:

$$\left(\frac{2^\nu k k' k'' \dots}{p}\right) = \left(\frac{u}{p}\right) = 1.$$

Now consider the simple factors of the odd number t , which we shall divide into two classes. The first class will comprise the prime divisors of one of these two forms, $8n + 1$, $8n + 7$, and the numbers belonging to it will be designated by $g, g', g'', \dots$; the second class will be composed of numbers $h, h', h'', \dots$ contained in these two forms, $8n + 3$, $8n + 5$. One first has:

$$t = gg'g'' \dots \times hh'h'' \dots,$$

and one will then conclude from the equation $p = t^2 + 2u^2$:

$$\left(\frac{2p}{g}\right) = 1, \quad \left(\frac{2p}{g'}\right) = 1, \quad \left(\frac{2p}{g''}\right) = 1, \quad \text{etc.}$$

$$\left(\frac{2p}{h}\right) = 1, \quad \left(\frac{2p}{h'}\right) = 1, \quad \left(\frac{2p}{h''}\right) = 1, \quad \text{etc.}$$

On the other hand, by known theorems one has:

$$\left(\frac{2}{g}\right) = 1, \quad \left(\frac{2}{g'}\right) = 1, \quad \left(\frac{2}{g''}\right) = 1, \quad \text{etc.}$$

$$\left(\frac{2}{h}\right) = -1, \quad \left(\frac{2}{h'}\right) = -1, \quad \left(\frac{2}{h''}\right) = -1, \quad \text{etc.}$$

Comparing these relations with the preceding ones, one finds:

$$\left(\frac{p}{g}\right) = 1, \quad \left(\frac{p}{g'}\right) = 1, \quad \left(\frac{p}{g''}\right) = 1, \quad \text{etc.}$$

$$\left(\frac{p}{h}\right) = -1, \quad \left(\frac{p}{h'}\right) = -1, \quad \left(\frac{p}{h''}\right) = -1, \quad \text{etc.}$$

The application of the law of reciprocity to these last relations gives:

$$\left(\frac{g}{p}\right) = 1, \quad \left(\frac{g'}{p}\right) = 1, \quad \left(\frac{g''}{p}\right) = 1, \quad \text{etc.}$$

$$\left(\frac{h}{p}\right) = -1, \quad \left(\frac{h'}{p}\right) = -1, \quad \left(\frac{h''}{p}\right) = -1, \quad \text{etc.}$$

Hence, on multiplying,

$$\left(\frac{gg'g'' \cdots \times hh'h'' \cdots}{p}\right) = \left(\frac{t}{p}\right) = \pm 1,$$

where the upper or lower sign must be taken according as the numbers $h, h', h'', \dots$ are even or odd in number. But it is easy to see, from the equation $t = gg'g'' \cdots \times hh'h'' \cdots$, that the first case will occur when t is of one of these forms, $8n+1, 8n+7$, and the second when t is contained in one of these, $8n+3, 8n+5$. Therefore

$$\left(\frac{t}{p}\right) = 1, \quad \text{when } t = 8n+1 \text{ or } 8n+7,$$

$$\left(\frac{t}{p}\right) = -1, \quad \text{when } t = 8n+3 \text{ or } 8n+5.$$

Let us return to the equation $t^2 + 2u^2 = p$ and put it in the form of a congruence:

$$t^2 \equiv -2u^2 \pmod{p}.$$

Raising both sides to the power $\frac{p-1}{4}$, one will have ($\frac{p-1}{4}$ being even):

$$t^{\frac{p-1}{2}} \equiv 2^{\frac{p-1}{4}} u^{\frac{p-1}{2}} \pmod{p},$$

or, what comes to the same thing, having proved that $\left(\frac{u}{p}\right) = 1$:

$$t^{\frac{p-1}{2}} \equiv 2^{\frac{p-1}{4}} \pmod{p}.$$

Thus one sees that ± 2 (the double sign may be put, since $p = 8n+1$) is or is not a biquadratic residue with respect to p according as one has $\left(\frac{t}{p}\right) = 1$ or $\left(\frac{t}{p}\right) = -1$. Comparing this result with what precedes, one has the theorem:

“ p designating a prime number $8n+1$, if one writes $p = t^2 + 2u^2$, I say that ± 2 will or will not be a biquadratic residue with respect to p according as t is of one of these forms, $8n+1, 8n+7$, or of one of these, $8n+3, 8n+5$.”

This is the first of the two theorems of Mr. Gauss mentioned in the preamble of this memoir. The second of these theorems is relative to the decomposition of the number p into two squares, and can easily be deduced from the one just established.

Let $p = \varphi^2 + \psi^2$ (where ψ is supposed divisible by 4) and equate this value of p with the one we have just considered. We shall thus have

$$p = t^2 + 2u^2 = \varphi^2 + \psi^2,$$

and, after transposing,

$$t^2 - \psi^2 = (t + \psi)(t - \psi) = \varphi^2 - 2u^2.$$

Since φ is odd, the greatest common divisor of φ and u will be odd; designate it by m and put $\varphi = m\varphi'$, $u = mu'$. Substitution of these values in the preceding equation changes it into

$$(t + \psi)(t - \psi) = m^2(\varphi'^2 - 2u'^2).$$

The odd number $t + \psi$ is thus composed of two factors E and K , the first of which divides m^2 , the second $\varphi'^2 - 2u'^2$; and the same is true of $t - \psi$, whose factors we denote by F and L , L being allowed to be negative. We therefore have

$$\begin{aligned} t + \psi &= EK, & t - \psi &= FL, \\ m^2 &= EF, & \varphi'^2 - 2u'^2 &= KL. \end{aligned}$$

It is easy to verify that the numbers E and F are relatively prime. Indeed, let δ be a prime divisor of E , and suppose that δ also divides F . The number δ would be a common divisor of $t + \psi$ and $t - \psi$, and consequently would divide the number t , which is the half-sum of the preceding two numbers. On the other hand, from the fact that δ is a prime divisor of E , it follows successively, having regard to the equations $EF = m^2$, $u = mu'$, that m^2 , m , and u are multiples of δ . The numbers t and u would therefore have the common divisor δ , and $p = t^2 + 2u^2$ would not be a prime number.

The product of the numbers E and F , which are relatively prime, being a square, each of them is also a square. Let $E = e^2$; then $t + \psi = e^2K$. The odd square e^2 is of the form $8n + 1$, and K , as an odd divisor of $\varphi'^2 + 2u'^2$ (where φ' and u' are relatively prime), is of one of these forms, $8n + 1$, $8n + 7$. Thus the number $t + \psi$ will itself be of one of the forms $8n + 1$, $8n + 7$. The number ψ , which we know to be divisible by 4, will be of the form $8n$ or of the form $8n + 4$. It follows from the preceding that, in the first case, t will be of one of these two forms, $8n + 1$, $8n + 7$, and in the second of one of these, $8n + 3$, $8n + 5$. Comparing this result with the preceding theorem, one has this other theorem:

“ p designating a prime number $8n + 1$, and having written $p = \varphi^2 + \psi^2$ (where ψ is supposed divisible by 4), ± 2 will or will not be a biquadratic residue with respect to p according as ψ is of the form $8n$ or of the form $8n + 4$.”

There is a third theorem suitable for deciding whether ± 2 is or is not a biquadratic residue relative to a prime number $p = 8n + 1$, which may be stated as follows:

“Having made, in any manner, $p = t^2 - 2u^2$, ± 2 will or will not be a biquadratic residue with respect to p according as t is of one of the forms $8n + 1$, $8n + 3$, or of one of these, $8n + 5$, $8n + 7$.”

We shall not stop to prove this theorem, which can be established directly and by considerations analogous to those on which the proof of the first of the two preceding theorems is founded. One can also deduce it from each of the preceding theorems almost as one has just passed from the first to the second.

We now pass to more general considerations. Let b be a prime number $4n + 3$, p a prime number $4n + 1$ capable of being put under the form $t^2 - bu^2$, and let us propose to decide whether $-b$ is or is not a biquadratic residue with respect to p . It is easy to see that, if p can be put under the form $t^2 - bu^2$, one can always do so in such a way that u is even, and consequently t odd¹. Let therefore $p = t^2 - bu^2$, and put $u = 2^\nu k k' k'' \dots$, where $k, k', k'', \dots$ are the simple odd factors of u . One concludes immediately from the preceding equation:

$$\left(\frac{p}{k}\right) = 1, \quad \left(\frac{p}{k'}\right) = 1, \quad \left(\frac{p}{k''}\right) = 1, \quad \text{etc.}$$

The application of the reciprocity law then gives, p being of the form $4n + 1$:

$$\left(\frac{k}{p}\right) = 1, \quad \left(\frac{k'}{p}\right) = 1, \quad \left(\frac{k''}{p}\right) = 1, \quad \text{etc.}$$

Multiplying these relations together and with the identical relation $\left(\frac{2^\nu}{p}\right) = \left(\frac{2^\nu}{p}\right)$, one has the result

$$(\alpha) \quad \left(\frac{u}{p}\right) = \left(\frac{2^\nu}{p}\right).$$

Now decompose the odd number t into its simple factors and divide these factors into two classes. Those of the first class will be designated by $g, g', g'', \dots$, and are such that

$$(\beta) \quad \left(\frac{-b}{g}\right) = 1, \quad \left(\frac{-b}{g'}\right) = 1, \quad \left(\frac{-b}{g''}\right) = 1, \quad \text{etc.}$$

As for those which form the second class and which we shall designate by $h, h', h'', \dots$, they are such that

$$(\beta') \quad \left(\frac{-b}{h}\right) = -1, \quad \left(\frac{-b}{h'}\right) = -1, \quad \left(\frac{-b}{h''}\right) = -1, \quad \text{etc.}$$

The product of all these numbers is equal to t , that is,

$$t = gg'g'' \dots \times hh'h'' \dots.$$

Inspection of the equation $t^2 - bu^2 = p$ gives the results

$$\left(\frac{-bp}{g}\right) = 1, \quad \left(\frac{-bp}{g'}\right) = 1, \quad \left(\frac{-bp}{g''}\right) = 1, \quad \text{etc.}$$

¹To prove that this is always possible, we shall show that, if one has $t^2 - bu^2 = p$, t being even, it is easy to deduce from the values of t and u other numbers t' (odd) and u' which likewise satisfy the equation $t'^2 - bu'^2 = p$. Let r and s be the least numbers such that $r^2 - bs^2 = 1$; I say that r will be even. Indeed, it is known that, if b designates a prime number $4n + 3$, the equation $\rho^2 - b\sigma^2 = \pm 2$ is always possible, and that, ρ and σ being supposed the least numbers which satisfy it, one has $r = \rho^2 \mp 1$, $s = \rho\sigma$ (Theory of Numbers, nos. 44, 45). From this, and from the fact that the numbers ρ and σ are evidently both odd, it follows that r is an even number. This being so, if one multiplies together the equations $r^2 - bs^2 = 1$, $t^2 - bu^2 = p$, one obtains $(rt + bsu)^2 - b(ru + st)^2 = p$, and one will easily see that $rt \pm bsu$ is an odd number.

$$\left(\frac{-bp}{h}\right) = 1, \quad \left(\frac{-bp}{h'}\right) = 1, \quad \left(\frac{-bp}{h''}\right) = 1, \quad \text{etc.}$$

Comparison of these relations with the preceding ones then gives

$$\left(\frac{p}{g}\right) = 1, \quad \left(\frac{p}{g'}\right) = 1, \quad \left(\frac{p}{g''}\right) = 1, \quad \text{etc.}$$

$$\left(\frac{p}{h}\right) = -1, \quad \left(\frac{p}{h'}\right) = -1, \quad \left(\frac{p}{h''}\right) = -1, \quad \text{etc.}$$

Applying now the reciprocity law, one obtains

$$\left(\frac{g}{p}\right) = 1, \quad \left(\frac{g'}{p}\right) = 1, \quad \left(\frac{g''}{p}\right) = 1, \quad \text{etc.}$$

$$\left(\frac{h}{p}\right) = -1, \quad \left(\frac{h'}{p}\right) = -1, \quad \left(\frac{h''}{p}\right) = -1, \quad \text{etc.}$$

Multiplying these relations together, one will have

$$\left(\frac{gg'g'' \cdots \times hh'h'' \cdots}{p}\right) = \left(\frac{t}{p}\right) = \pm 1,$$

the upper or lower sign holding according as the numbers $h, h', h'', \dots$ are even or odd in number. On the other hand, if one applies the reciprocity law to the relations (β) and (β') , one obtains, b being of the form $4n + 3$,

$$\left(\frac{g}{b}\right) = 1, \quad \left(\frac{g'}{b}\right) = 1, \quad \left(\frac{g''}{b}\right) = 1, \quad \text{etc.}$$

$$\left(\frac{h}{b}\right) = -1, \quad \left(\frac{h'}{b}\right) = -1, \quad \left(\frac{h''}{b}\right) = -1, \quad \text{etc.}$$

equalities which, when multiplied together, give

$$\left(\frac{gg'g'' \cdots \times hh'h'' \cdots}{b}\right) = \left(\frac{t}{b}\right) = \pm 1,$$

where the upper or lower sign must be taken according as the numbers $h, h', h'', \dots$ are even or odd in number.

Comparison of this result with the one obtained a moment ago shows that one always has

$$(\gamma) \quad \left(\frac{t}{p}\right) = \left(\frac{t}{b}\right).$$

Let us now return to the equation $p = t^2 - bu^2$ and put it in congruence form:

$$t^2 \equiv bu^2 \pmod{p}.$$

From this, raising both sides to the power $\frac{p-1}{4}$, one obtains

$$t^{\frac{p-1}{2}} \equiv b^{\frac{p-1}{4}} u^{\frac{p-1}{2}} \pmod{p}.$$

The formula (α) , namely $\left(\frac{u}{p}\right) = \left(\frac{2^\nu}{p}\right)$, where 2^ν denotes the highest power dividing u , can be presented in another manner. For this it is necessary to distinguish two cases, according as p is of the form $8n + 1$ or of the form $8n + 5$.

If p is a prime number $8n+1$, one has, as is known, $\left(\frac{2}{p}\right) = 1$, and consequently $\left(\frac{2^\nu}{p}\right) = 1$; the formula in question therefore becomes in this case $\left(\frac{u}{p}\right) = 1$. If p is a prime number $8n+5$, the number ν is equal to 1; for if ν were greater than unity, u^2 would be divisible by 8, and $p = t^2 - bu^2$ would be of the form $8n+1$. Moreover, since in this case $\left(\frac{2}{p}\right) = -1$, formula (α) becomes $\left(\frac{u}{p}\right) = -1$. The two cases just examined are included in the formula

$$\left(\frac{u}{p}\right) = (-1)^{\frac{p-1}{4}},$$

which is equivalent to the congruence

$$u^{\frac{p-1}{2}} \equiv (-1)^{\frac{p-1}{4}} \pmod{p}.$$

Substituting the value which this gives for $u^{\frac{p-1}{2}}$ in the congruence obtained above, one has

$$t^{\frac{p-1}{2}} \equiv (-b)^{\frac{p-1}{4}} \pmod{p},$$

a result which shows that $-b$ will or will not be a biquadratic residue with respect to p according as t is or is not a quadratic residue with respect to p . Comparing this result with that contained in formula (γ) , one arrives at the theorem which we shall state:

“Let b designate a prime number $4n+3$, and p a prime number $4n+1$ capable of being put under the form $t^2 - bu^2$. Having written $p = t^2 - bu^2$ (where t is supposed odd), I say that $-b$ will or will not be a biquadratic residue with respect to p according as t is or is not a quadratic residue with respect to b .”

4.

We shall now deduce from this theorem another by means of which one can decide still more quickly whether $-b$ is or is not a biquadratic residue with respect to p . Preserve the preceding notations and put $p = \varphi^2 + \psi^2$ (where ψ is supposed even). Equating this value of p to the one considered above, one has

$$p = t^2 - bu^2 = \varphi^2 + \psi^2,$$

and, after transposing,

$$t^2 - \psi^2 = (t + \psi)(t - \psi) = \varphi^2 + bu^2.$$

There are now two cases to distinguish, according as φ is or is not divisible by b . We begin with the latter of these two cases. Let m be the greatest common divisor of φ and u , which will be odd (φ being odd) and not divisible by b , and put $\varphi = m\varphi'$, $u = mu'$. Substitution of these values in the last equation changes it into

$$(t + \psi)(t - \psi) = m^2(\varphi'^2 + bu'^2).$$

It is evident, from this equation, that $t + \psi$ is composed of two factors E and K , the first of which divides m^2 , the second $\varphi'^2 + bu'^2$, and that the same is true of $t - \psi$. Denoting the factors of this last number by F and L , we shall have the equations

$$\begin{aligned} t + \psi &= EK, & t - \psi &= FL, \\ m^2 &= EF, & \varphi'^2 + bu'^2 &= KL. \end{aligned}$$

RESEARCHES ON THE PRIME DIVISORS OF A CLASS OF FORMULAE OF THE FOURTH DEGREE.

First memoir. Continuation: printed pages 75-80.

It is very easy to make sure that the numbers E and F are relatively prime. Indeed, let δ be any simple factor of E (which will necessarily be odd, E itself being odd), and suppose that δ also divides F . Inspection of the preceding equations shows that, under this supposition, δ would be a common divisor of $t + \psi$ and $t - \psi$, and consequently would divide the number t , which is the half-sum of the preceding two. But on the other hand one sees that δ , as a prime divisor of E , divides m^2 , and consequently m and u (u being $= mu'$). The numbers t and u would therefore, under this supposition, have the common factor δ , which is absurd, since $p = t^2 - bu^2$ is a prime number. It is therefore proved that E and F cannot have a common divisor; and since the product of these numbers is a square, each of them is likewise a square. Thus one has:

$$\left(\frac{E}{b}\right) = 1, \quad \left(\frac{F}{b}\right) = 1.$$

As for the numbers K and L , they are likewise such that one has:

$$\left(\frac{K}{b}\right) = 1, \quad \left(\frac{L}{b}\right) = 1.$$

Indeed, one sees from the last equation (p. 74) that these numbers divide, the one and the other, the formula $\varphi'^2 + bu'^2$ (where φ' and bu' are relatively prime); and it is known, by a known theorem that follows easily from the law of reciprocity (*Théorie des Nombres*, no. 197), that every odd divisor R of such a formula is such that $\left(\frac{R}{b}\right) = 1$. Multiplying the first of the last relations by the third, and the second by the fourth, one obtains:

$$\left(\frac{EK}{b}\right) = 1, \quad \left(\frac{FL}{b}\right) = 1,$$

or, which is the same thing:

$$(\delta) \quad \left(\frac{t + \psi}{b}\right) = 1, \quad \left(\frac{t - \psi}{b}\right) = 1.$$

Examining in a similar way the case in which φ is divisible by b , one will find that, in this case, one of the numbers $t + \psi$ and $t - \psi$ is divisible by b , and that the other is, as in the case with which we have just been occupied, a quadratic residue with respect to b ; so that, denoting by $t \pm \psi$ that one of these numbers which is divisible by b , one has:

$$(\delta') \quad t \pm \psi \equiv 0 \pmod{b}, \quad \left(\frac{t \mp \psi}{b}\right) = 1.$$

It follows from the theorem stated at the end of the preceding paragraph that, in order to decide whether $-b$ is or is not a biquadratic residue with

respect to p , everything reduces to the question of knowing whether t is or is not a quadratic residue with respect to b . We can now show, by means of the results just obtained, that this latter question can be decided without knowing t , that is, without its being necessary to put p under the form $t^2 - bu^2$. Suppose first that φ is not divisible by b . The relations (δ) , which hold in this case, can be presented in this way, observing that b is a prime number $4n + 3$:

$$\left(\frac{\psi + t}{b}\right) = 1, \quad \left(\frac{\psi - t}{b}\right) = -1.$$

One immediately draws from the equation $p = t^2 - bu^2$: $t^2 \equiv p \pmod{b}$, a result which shows that, on solving the congruence $\chi^2 \equiv p \pmod{b}$, and denoting by χ one of its roots taken at random, one will have $t \equiv \chi$, or $t \equiv -\chi \pmod{b}$. In the first case, one has:

$$\left(\frac{t}{b}\right) = \left(\frac{\chi}{b}\right), \quad \left(\frac{\chi + \psi}{b}\right) = 1,$$

for it is evident that the expressions $\left(\frac{t}{b}\right)$ and $\left(\frac{t+\psi}{b}\right)$ do not change when one puts, in place of t , another number χ which differs from t only by a multiple of b . Multiplying the last relations together, one obtains:

$$\left(\frac{t}{b}\right) = \left(\frac{\chi(\chi + \psi)}{b}\right).$$

Consider now the other case, in which $t \equiv -\chi \pmod{b}$. One will then have:

$$\left(\frac{t}{b}\right) = -\left(\frac{\chi}{b}\right), \quad \left(\frac{\psi + \chi}{b}\right) = -1,$$

the last of these relations resulting from the formula $\left(\frac{\psi-t}{b}\right) = -1$, if one puts, in place of $-t$, the number χ , which differs from it only by a multiple of b . Multiplying the preceding relations together, one will find, as above:

$$\left(\frac{t}{b}\right) = \left(\frac{\chi(\chi + \psi)}{b}\right).$$

We have therefore arrived at this remarkable result: if φ is not divisible by b , one can decide very simply whether one has:

$$\left(\frac{t}{b}\right) = 1 \quad \text{or} \quad \left(\frac{t}{b}\right) = -1.$$

It will suffice to seek a number satisfying the congruence $\chi^2 \equiv p \pmod{b}$; having found such a number χ , one will always have:

$$\left(\frac{t}{b}\right) = \left(\frac{\chi(\chi + \psi)}{b}\right).$$

Let us now come to the case in which φ is divisible by b . The relations (δ') which then hold are:

$$t \pm \psi \equiv 0 \pmod{b}, \quad \left(\frac{t \mp \psi}{b}\right) = 1.$$

In the last of these relations one may add to $t \mp \psi$ any multiple of b . If one adds to it the number $t \pm \psi$, which, by virtue of the first relation, is divisible by b , there will result $\left(\frac{2t}{b}\right) = 1$, a result which entails this other one: $\left(\frac{t}{b}\right) = \left(\frac{\psi}{b}\right)$.

Combining this result with a known theorem, one will see that, in the case of φ divisible by b , one has $\left(\frac{t}{b}\right) = 1$ or $\left(\frac{t}{b}\right) = -1$, according as t is of the form $8n + 7$ or of this one: $8n + 3$.

Comparing what has just been proved with the theorem established at the end of the preceding paragraph, one will see that it is possible to decide, independently of the knowledge of the number t , whether $-b$ is or is not a biquadratic residue with respect to p . Since the result thus reached no longer contains any trace of the number t , one is led to believe that it does not presuppose the possibility of the equation $t^2 - bu^2 = p$, from which the number t takes its origin, and that it is generally true for all values of b and p such that $\left(\frac{b}{p}\right) = 1$. This is indeed what takes place, as one can prove, by examining, instead of the equation $t^2 - bu^2 = p$, the more general equation $t^2 \pm Mu^2 = p$, where M denotes the product of any number of distinct prime numbers, and then combining the results of this examination with the following theorem, of whose truth one cannot doubt, but whose rigorous demonstration nevertheless presents difficulties:

“ c denoting any prime number whatever and p a prime number $4n + 1$, such that $\left(\frac{c}{p}\right) = 1$, one can always determine a number δ , composed of simple factors all smaller than c , and such that the equation $t^2 \pm c\delta u^2 = p$ is soluble.”

However that may be, one may remark that the theorem which we are about to state is rigorously proved, by what precedes, for all values of b such that the formula $t^2 - bu^2$ has only the single quadratic divisor $\pm(t^2 - bu^2)$. Casting one's eyes over the first of the tables added to the *Théorie des Nombres*, one sees that all the prime numbers $4n + 3$ smaller than 136 (the limit of the table) are in this case, excepting the single number 79. It would be easy, by following the path just indicated, to prove the truth of the theorem for this value or for any other particular value of b , whatever p may be; but in order to establish it in all its generality, it is necessary first to prove, as we have already said, that one can always satisfy the condition which we stated only a moment ago.

Theorem I.

“ b denoting a prime number $4n + 3$, and p a prime number $4n + 1$ such that $\left(\frac{b}{p}\right) = 1$, if one puts $p = \varphi^2 + \psi^2$ (where ψ is supposed even), one will have the following rule for deciding whether $-b$ is or is not a biquadratic residue with respect to p : if φ is divisible by b , $-b$ will or will not be a biquadratic residue according as b is of the form $8n + 7$ or of this one: $8n + 3$. If φ is not divisible by b , one will seek a number χ such that $\chi^2 \equiv p \pmod{b}$. This being done, $-b$ will or will not be a biquadratic residue with respect to p , according as one has $\left(\frac{\chi(\chi+\psi)}{b}\right) = 1$ or $\left(\frac{\chi(\chi+\psi)}{b}\right) = -1$.”

By successively taking $b = 3$, $b = 7$, etc., one will obtain the following particular theorems, which are analogous to the second of Mr. Gauss's theorems and which must be regarded as rigorously proved by the considerations that we have set out in this paragraph and in the preceding one.

" p denoting a prime number $12n + 1$, if one puts $p = \varphi^2 + \psi^2$ (where ψ is supposed even), one of the numbers φ and ψ will necessarily be divisible by 3. This being done, I say that -3 will or will not be a biquadratic residue with respect to p , according as it is ψ or φ which is divisible by 3."

" p denoting a prime number of one of these forms: $28n + 1$, $28n + 9$, $28n + 25$, if one puts $p = \varphi^2 + \psi^2$, I say that -7 will be a biquadratic residue with respect to p if one of the numbers φ and ψ is divisible by 7, and that -7 will be a biquadratic non-residue with respect to p if neither of these numbers is divisible by 7."

etc.

5.

Let us denote by a a prime number $4n + 1$, and by p a prime number likewise $4n + 1$, and moreover capable of being put under the form $t^2 - au^2$. We can therefore put $p = t^2 - au^2$, where the number t , as is easy to see, is necessarily odd. One will find, as in paragraph 3, that if one denotes by 2^ν the highest power of 2 that divides u , one has $\left(\frac{u}{p}\right) = \left(\frac{2^\nu}{p}\right)$, a relation which one will then change, as in the cited place, into this one: $\left(\frac{u}{p}\right) = (-1)^{\frac{p-1}{4}}$.

Let us now decompose the odd number t into its simple factors, putting $t = gg'g'' \dots \times hh'h'' \dots$, where the prime numbers g, g', g'' , etc., h, h', h'' , etc. are such that:

$$(\varepsilon) \quad \begin{cases} \left(\frac{-a}{g}\right) = 1, & \left(\frac{-a}{g'}\right) = 1, & \left(\frac{-a}{g''}\right) = 1, & \text{etc.} \\ \left(\frac{-a}{h}\right) = -1, & \left(\frac{-a}{h'}\right) = -1, & \left(\frac{-a}{h''}\right) = -1, & \text{etc.} \end{cases}$$

The equation $t^2 - au^2 = p$ gives immediately:

$$\begin{aligned} \left(\frac{-ap}{g}\right) &= 1, & \left(\frac{-ap}{g'}\right) &= 1, & \left(\frac{-ap}{g''}\right) &= 1, & \text{etc.} \\ \left(\frac{-ap}{h}\right) &= 1, & \left(\frac{-ap}{h'}\right) &= 1, & \left(\frac{-ap}{h''}\right) &= 1, & \text{etc.} \end{aligned}$$

Comparing these relations with the preceding ones, one obtains:

$$\begin{aligned} \left(\frac{p}{g}\right) &= 1, & \left(\frac{p}{g'}\right) &= 1, & \left(\frac{p}{g''}\right) &= 1, & \text{etc.} \\ \left(\frac{p}{h}\right) &= -1, & \left(\frac{p}{h'}\right) &= -1, & \left(\frac{p}{h''}\right) &= -1, & \text{etc.} \end{aligned}$$

The application of the law of reciprocity will then give, p being of the form $4n + 1$:

$$\begin{aligned} \left(\frac{g}{p}\right) &= 1, & \left(\frac{g'}{p}\right) &= 1, & \left(\frac{g''}{p}\right) &= 1, & \text{etc.} \\ \left(\frac{h}{p}\right) &= -1, & \left(\frac{h'}{p}\right) &= -1, & \left(\frac{h''}{p}\right) &= -1, & \text{etc.} \end{aligned}$$

whence one concludes, by multiplying:

$$\left(\frac{gg'g'' \dots \times hh'h'' \dots}{p}\right) = \left(\frac{t}{p}\right) = \pm 1,$$

the upper or lower sign taking place according as the prime numbers h, h', h'' , ... are even or odd in number. One can state this result in a slightly different way, by saying that $\left(\frac{t}{p}\right)$ is equal to the product of the second members of the relations (ε) .

In the relations (ε) , one may everywhere change $-a$ into a , provided that at the same time one changes the sign of the second members of those of these relations in which there occurs a prime number $g, g', g'', \dots, h, h', h'', \dots$ of the form $4n + 3$. One will thus have:

$$(\zeta) \quad \left\{ \begin{array}{l} \left(\frac{a}{g}\right) = \pm 1, \quad \left(\frac{a}{g'}\right) = \pm 1, \quad \left(\frac{a}{g''}\right) = \pm 1, \quad \text{etc.} \\ \left(\frac{a}{h}\right) = \pm 1, \quad \left(\frac{a}{h'}\right) = \pm 1, \quad \left(\frac{a}{h''}\right) = \pm 1, \quad \text{etc.} \end{array} \right.$$

Comparing the second member of each of these relations with the second member of the corresponding relation of table (ε) , one will find as many changes of sign as there are prime numbers $4n + 3$ among the numbers $g, g', g'', \dots, h, h', h'', \dots$.

The number of changes will therefore be even when among the numbers $g, g', g'', \dots, h, h', h'', \dots$ there is an even number of the form $4n + 3$. In this case, which will occur every time that $t = gg'g'' \dots \times hh'h'' \dots$ is of the form $4n + 1$, the product of the second members of the relations (ζ) will therefore be the same as the product of the second members of the relations (ε) .

On the other hand, one may, by virtue of the law of reciprocity, reverse the first members of the relations (ζ) without its being necessary to change the sign of any of the second members of these relations. The product of all the reversed expressions, that is, $\left(\frac{gg'g'' \dots \times hh'h'' \dots}{a}\right)$ or $\left(\frac{t}{a}\right)$, is therefore equal to the product of the second members of the relations (ζ) and consequently also, according to what has previously been seen, equal to the product of the second members of the relations (ε) . Now, having proved above that this last product is equal to $\left(\frac{t}{p}\right)$, one has:

$$\left(\frac{t}{p}\right) = \left(\frac{t}{a}\right).$$

This result concerns the case in which t is of the form $4n + 1$; if one examines in a similar way the case of $t = 4n + 3$, one will find that one then has:

$$\left(\frac{t}{p}\right) = - \left(\frac{t}{a}\right).$$